

Optimal Trading by a “Large Shareholder”[†]

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ABSTRACT.

A large shareholder who undertakes costly effort to improve a firm’s dividends faces a tradeoff. Selling shares will likely lower the share price (as the market anticipates a reduction in effort), while holding the shares implies a less diversified investment portfolio. Moreover, in a dynamic setting a time-consistency problem emerges: once some shares are sold, the incentive to sell additional shares may increase since the large shareholder is less exposed to the resultant price declines. We analyze a multi-period model for the optimal trading strategy of a large shareholder. We consider the case in which the large shareholder can commit to a trading strategy, and the case in which such commitment is impossible. Absent commitment, the problem is similar to durable goods monopoly: the share price today depends on the shares expected to be sold in the future. We demonstrate that an analog of the “Coase conjecture” holds in the sense that shareholdings ultimately converge to the competitive outcome of efficient diversification. Unlike the standard monopoly setting, however, this outcome is suboptimal in that there is inefficient monitoring. We provide conditions for which convergence to this outcome is gradual, and when it occurs immediately. In the continuous trading approximation, our results produce a simple formula for the equilibrium share price in this setting: the trading strategy of the large shareholder can be ignored, and today’s share price is simply the present value of dividends given *constant* holdings by the large shareholder, but adjusted by a risk premium that reflects the large shareholder’s (rather than investors’) risk aversion. We demonstrate that our model provides a rational for both IPO underpricing and the use of lockup provisions. Finally, we generalize our results outside the moral hazard framework.

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1. Introduction

Corporate governance policies can dramatically impact performance of a company or even the economy as a whole. One of the hotly debated issues of corporate finance as well as political economics is the issue of optimal structure and concentration of corporate ownership. In the several emerging markets of Eastern Europe, concentrated ownership is often seen as the “magic bullet” which could facilitate the creation of more motivated company leaderships and, ultimately, better performing economies. Others argue that U.S. style of relatively dispersed ownership may be preferable because it facilitates more market liquidity and, thus, improves the efficiency of the market.

Considerable progress has been achieved in developing models of optimal ownership structure. One weakness of existing models, however, is that the vast majority of them are static in nature.¹ In reality, ownership structure of companies changes over time. This is particularly easy to observe for firms going public. Mikkelson et al (1997) show that, for U.S. companies which went public in 1980-1983 period, the median ownership stake of the top management falls from 68%, prior to IPO, to 44%, immediately after the IPO. Five years after the IPO, median ownership stake of the top management is 29%, while ten years after going public it falls further to 18%. A decrease in ownership concentration is often observed for the individual large shareholders/managers as well. Indeed, in the year 2000 Bill Gates disposed of at least 65 million shares of Microsoft stock (roughly 8.3% of his holdings) in 12 separate transactions. In the period from 1992 to 1999, he decreased his ownership stake from 30.1% to 15.3%.

Large shareholders/managers may decrease their stake in the company in order to capture diversification benefits and spread their consumption over time. For the investors in such companies, it is important to correctly anticipate the large shareholder’s trading policy. First, following Jensen and Meckling (1976), managerial incentives are better aligned with the interests of shareholders the larger the stake managers have in the company. Therefore, changes in the large shareholder’s stake are likely to influence management decisions, which, in turn, could impact the company’s valuation.² A second, generally smaller, effect is that by selling shares the shareholder forces investors to hold more of the risk of the firm, which may raise the market risk premium for the shares. Since selling off his shares would likely lower the company share price, a large shareholder/manager is typically facing a trade-off between a lower share price and a less diversified personal investment portfolio and/or less consumption.³ Another reason for diluting the ownership could be the need for market liquidity (see Section 7).⁴

¹ An important critique of static models of corporate governance as applied to the economic transition can be found in Roland (2000).

² For example, the *International Herald Tribune* reported on August 22, 2001 that shares of Tiscali, Europe’s No. 2 internet company, rose sharply after they announced that their two largest shareholders would refrain from selling off a substantial part of their stakes. (We thank G. Cespa for this example.)

³ Here we neglect the potential negative effects of managerial entrenchment which can lead to non-monotonic dependence between the managerial ownership stake and the company valuation (see Stultz (1988)).

⁴ In this paper we assume away informational asymmetry between the large shareholder and other investors. That precludes market manipulations by the large shareholder as a motive for trade.

An outside large shareholder is facing a similar problem. Jensen (1989) argues that diffusely held firms should be worth less than ones with concentrated ownership, *ceteris paribus*: since no individual investor would find it worthwhile to engage in monitoring, managers would shirk.⁵ Again, the large outside shareholder is facing the trade-off between the benefits of monitoring, which is worth undertaking only if she owns a sufficient share of the company stock, and a loss of diversification. Likewise, investors need to anticipate the future holdings of the large shareholder to correctly anticipate the performance and value of the firm. Selling shares will likely lower the share price (as the market anticipates a reduction in effort), while holding the shares implies a less diversified investment portfolio. Moreover, in a dynamic setting a time-consistency problem emerges: once some shares are sold, the incentive to sell additional shares may increase since the large shareholder is less exposed to the resultant price declines. As a result, large shareholders may be inclined to decrease their position over time. Venture capitalists display just such a behavior by always exiting their portfolio companies prior to the dissolution of the venture partnership.⁶

In this paper, we analyze the optimal trading and consumption policy of a shareholder/manager of a publicly traded company who is “large” in the following sense:⁷ Her stake in the company determines her optimal level of effort, which affects, in turn, the company’s future dividends and, therefore, the stock price. Given that the firm is not infinitesimal, if other investors are risk averse, the large shareholder’s trading policy can also affect the market risk premium for the risk of the stock.⁸ Her trading policy, however, does not influence the prices of other goods in the economy and in that respect the firm acts as a competitive firm.

In an economy where the large shareholder undertakes monitoring effort that affects the mean and variance of normally distributed dividends, and both the large shareholder and the infinitesimal investors have CARA preferences, we formulate and solve a multi-period model for the optimal trading and consumption policy of the large shareholder, as well as the equilibrium share price. We consider the case in which the large shareholder can commit to a trading strategy, and the case in which such commitment is impossible. When commitment is possible, our results generalize several previously known results (see Section 1.1). Noting that the commitment strategy is time-inconsistent, we show that the time-consistent strategy of a large shareholder is to sell off the initial stake in the company gradually over time, asymptotically approaching the competitive level. In the continuous time limit, we obtain a closed form solution for the equilibrium share price in terms of the primitives of the model, and a differential equation for the optimal share holdings. The solution for the equilibrium share price is strikingly simple: the trading

⁵ In contrast, Burkart et al (1997) argue that a very large outside ownership stake, and the corresponding large level of monitoring, can be detrimental to the manager’s incentive to take initiative. In parallel with Stulz (1988) for the insiders’ holdings, they predict non-monotonic dependence of the company stock as a function of the outside ownership stake.

⁶ A typical lifespan of an American venture fund is ten years but most portfolio companies are exited prior to that date. For the reasons and modes of VC exits see Cumming and MacIntosh (2001) and references therein.

⁷ Throughout the paper we, for definiteness, adopt the large shareholder/manager terminology. With the appropriate re-interpretation, however, the model applies to an external large shareholder as well.

⁸ All of the major conclusions remain valid if we assume that the investors are risk-neutral.

strategy of the large shareholder can be ignored, and today's share price is simply the present value of dividends given constant holdings by the large shareholder, but adjusted by a risk premium that reflects the large shareholder's (rather than investors') risk aversion. Our paper generalizes the existing literature, in addition, by allowing the large shareholder to influence dividend volatility in addition to the mean.

Having solved the model, we then apply it to a number of interesting issues. In Section 6 we show that a large stake can be created in a natural way in the process of an Initial Public Offering and that post-IPO large shareholder dynamics plays an important role in determining an optimal initial stake. We also show that our model provides a rationale for IPO underpricing. In addition, we study the effect that a lockup provision has on the large shareholder's welfare, the share price, and short and long run ownership concentration. In particular, we show that if the environment is stationary, the agent's welfare and the share price is enhanced by a lockup restriction, *ceteris paribus*. Finally, we have the empirical predictions that the speed of adjustment of the large shareholder's stake (post-lockup) should be positively related to the presence of a lockup restriction, and negatively correlated with degree of IPO underpricing.

Since company shares are "durable," a useful analogy can be drawn between our model and the durable goods monopolist problem.⁹ A monopolist in a dynamic durable good market faces an erosion of his market power, since her prior sales face competition from her future sales. Coase (1972) conjectures that, as a result, the price that the monopolist can charge would drop to the competitive level. Our paper demonstrates the way in which such adjustment occurs: in the limit of continuous trading, the adjustment is gradual when the marginal value of a share to the large shareholder is increasing in the large shareholder's stake, and immediate when the marginal value is decreasing.

In the last section of the paper, we demonstrate how our results can be generalized to include durable goods monopoly in the presence of a competitive rental market. Our results generalize the previous literature by not restricting marginal cost to be constant. We provide an explicit characterization of equilibrium price dynamics in the case of a general cost structure.

The paper is organized as follows. In the rest of this section we discuss the related literature. In Section 2 we describe the model. In Section 3 we perform some preliminary analysis culminating, in Section 3.4, with the analysis of the optimal trading and consumption policy in the commitment case. In Section 4 we discuss optimal trading strategy without commitment in the discrete time setting. Section 5 shows that the results of Section 4 dramatically simplify in the continuous time limit. In Section 6 we show that our model provides a rationale for both IPO underpricing and the use of lockup provisions. Section 7 extends our analysis to include other interesting possibilities. We conclude with suggestions for future research.

1.1. Related Literature

An important feature of our paper is that it links some traditionally separated branches of literature: asset pricing in the presence of a "large shareholder", asset pricing in the

⁹ See Bizer and DeMarzo (1993) for a discussion of this relationship.

presence of an optimal monitoring / moral hazard problem, corporate governance literature on large shareholders, and the Principal-Agent problem. In addition, an important parallel exists with the durable goods monopolist problem. Below we briefly explore the connection between these literatures in relation to our problem.

Asset pricing in the presence of a “large shareholder.” Cvitanic (1997), Cuoco and Cvitanic (1998), and El Karoui et al (1997) consider the portfolio optimization problem of a large investor when the drift rate of the risky and riskless assets have an exogenously specified dependence on the large investor’s holdings. Use of martingale methodology leads them to preclude the possibility of the large shareholder’s impact on the stock price volatility. Since these papers do not impose equilibrium market clearing, the price process is exogenously specified.¹⁰ In contrast, we explicitly model the moral hazard problem relating both the mean and volatility of the dividend process to shareholdings. In addition, we impose market clearing and endogenously derive the equilibrium market price.

Basak (1996) does consider an equilibrium model. However, there is no moral hazard and the agent’s position does not affect dividends. Instead, Basak solves for the continuous-time commitment strategy of an expected utility maximizing agent who behaves as a price-setter rather than a price-taker. That is, the agent recognizes that his portfolio choices influence state prices, including, e.g., the risk-free rate in the economy. This is because the agent’s holdings determine the aggregate position that must be held by the residual, competitive sector of the economy.¹¹ The paper makes note of time-inconsistency of the commitment strategy, but does not address it explicitly. Our model, in contrast, includes moral hazard and directly considers both commitment and time-consistent policies.

The closest to our approach in this literature is Kihlstrom (1998). He considers a three-period model similar to Basak’s but solves, in addition, for the time-consistent strategy of the large shareholder. He shows that the problem exhibits Coasian dynamics, in the sense that the agent’s portfolio in the second period is closer to the competitive outcome than it would be if the agent could commit to a trading strategy at the start. Like Basak, he does not include moral hazard and so the agent’s holdings do not affect dividends. While our set-up allows for greater tractability in a multi-period and, especially, in a continuous time setting, and incorporates the manager’s moral hazard problem, Kihlstrom (1998) contains many of the essential insights into the dynamics of the problem.

Asset pricing in the presence of a moral hazard. Kihlstrom and Matthews (1990), and Magill and Quinzii (2000) develop a static *competitive* asset pricing theory in the presence of moral hazard. In these models, the manager’s stake in the company influences her optimal effort choice, and she recognizes the price impact of her trades due to investors’ expectations of her change in effort after trading. However, she does not consider the effect of her trades on equilibrium risk premia (i.e., she is a price-taker in that regard). Since these are static models, time-inconsistency is not an issue. In our model, the agent considers the price impact of trades due to both the moral hazard and

¹⁰ Jarrow (1992) also exogenously specifies a price path and studies market manipulations by the large shareholder.

¹¹ Related static models are Lindenberg (1979) and Grinblatt and Ross (1985).

risk-sharing considerations. In addition, we focus on the dynamic trading case. We show, however, that our commitment case contains as a special case some of the results of Kihlstrom and Matthews (1990) (see Section 3.4, fn. 38).¹²

Monitoring by large shareholders (corporate governance). Ever since the influential work by Berle and Means (1932), separation between the corporate ownership and control and the related issue of incentive alignment have been intensely scrutinized in the literature on corporate governance.¹³ For our purposes, most relevant is the branch of that literature related to optimal monitoring by large shareholders. Most of the existing literature on optimal monitoring belongs to the class of static models.¹⁴ The closest to our approach are Admati et al (1994) and Stoughton and Zechner (1998) who focus on the risk-sharing/monitoring benefits trade-off. Like our paper, Admati et al (1994) consider optimal ownership policy of a large shareholder after the IPO. In essence, in the commitment case, their model is a static version of our commitment case (though they also allow for multiple risky assets). In the absence of commitment, their model allows repeated trading rounds prior to a *single* monitoring decision and cash flow. They note the time consistency problem when such repeated trading is allowed. Rather than consider a fully dynamic model with multiple rounds of monitoring and cash flows, they define a static equilibrium concept of “global stability” as a portfolio from which the large shareholder has no desire to trade. The “globally stable” allocation exists in their setting when marginal benefits of the large shareholder are decreasing in her company stake, and coincides with the price-taking (Walrasian) equilibrium. We re-derive that result in a more general setting and show that, in that case, in continuous time the adjustment to this allocation is *immediate*. On the other hand, when marginal benefits of the large shareholder are increasing in the company stake, they find that no globally stable equilibrium exists. In contrast, we show that when trading, cash flows and monitoring occur continuously, an equilibrium does exist, and involves a *gradual* adjustment towards the price-taking solution. This allows us to determine the effect on, and the dynamics of, equilibrium share prices. While they do not focus on the question of the origin of a large shareholder, Admati et al (1994) note that, for an outside shareholder, there is cost of being “large” due to the free riding of small investors, but that there is also a potential benefit in the form of reduced transaction costs.

Stoughton and Zechner (1998) use a closely-related model in order to motivate the creation of a large shareholder stake at the time of an IPO, as well as to explain the well-known effects of IPO underpricing and share rationing. The idea there is that an entrepreneur, at the time of the IPO, sells off her entire stake with an aim of maximizing her proceeds from the sale. In order to ensure a sufficient level of monitoring, the underwriter attracts a large institution to acquire a sufficiently large stake by either offering it a price discount vis-à-vis small investors (price discrimination), or charging a uniform price and partially restricting the small investors from acquiring as many shares as they would like (IPO underpricing and share rationing). In fact, our model can be

¹² A number of static asset pricing models attempt to use moral hazard in order to explain the equity premium puzzle (see Kocherlakota (1998) and the references therein).

¹³ See Shleifer and Vishny (1997) for a review of that literature.

¹⁴ A vast majority of the models in this group assume risk-neutrality of the agents in the economy: with no incentive to trade, agents do not deviate from the equilibrium.

interpreted as a fully-fledged dynamic version of their underpricing model (see Section 5.2).

Instead of focusing on the risk sharing/monitoring trade-off, Bolton and von Thadden (1998), Kahn and Winton (1998), and Maug (1998), among others, consider the trade-off between the liquidity benefits of a dispersed ownership and control benefits of a concentrated ownership.¹⁵ Their asymmetric information models are part of an extensive branch of literature in corporate governance that aims to shed light on how and why a large shareholder position would be created in the first place.

We do not attempt to be too specific as to what may cause the initial creation of the large stake (though see our discussion of IPO's in Section 6). The easiest explanation would be to think of the large shareholder either as a top manager or a venture capitalist in a firm recently going public. Burkart et al (1998), demonstrate that takeovers can lead to the creation of a large shareholder as well as to the emergence of the additional agency problems.¹⁶ While we do not explicitly consider the possibility of a takeover, one can think of our model as, alternatively, describing a post-takeover situation where the large shareholder is the new majority stake owner of the company.

An open issue of considerable interest is to incorporate the possibility of a takeover into a dynamic framework like ours.

Principal-agent literature. Under certain conditions, our commitment model coincides with the optimal contract for the Principal-Agent problem studied by Holmstrom and Milgrom (1987), who show that the linear contract is optimal among all contracts (see Section 3.4, fn. 37 for more details). One interpretation of our model is that we consider the consequences of allowing the Agent to trade shares of the firm. In that sense our paper fits into the literature on Principal-Agent models when the Agent has access to financial markets (see, e.g., Bizer and DeMarzo (1999)).

Durable Goods Monopoly and the Coase Conjecture. Coase (1972) considered the problem faced by a durable goods monopolist and argued that, absent commitment, the monopolist's ability to sell repeatedly will force him to charge the competitive price. This idea was formalized in a series of papers (e.g., Stokey (1981), Gul et al (1985), Ausubel and Deneckere (1989)) confirming the conjecture when the monopolist has constant marginal cost, a finite horizon, and trading opportunities are frequent. DeMarzo and Bizer (1993) established the connection to securities markets when asset payoffs depend upon portfolio allocations. In particular, they consider a model of a large shareholder who engages in monitoring or who may facilitate a takeover, as well as model of debt buybacks.¹⁷ That paper, however, restricts attention to cases in which

¹⁵ Starting from Kyle (1985), there are many papers discussing optimal dynamic liquidity trading by large shareholders including, recently, Vayanos (2000), Chau and Vayanos (2001), Huberman and Stanzl (2001), among others. In this paper we abstract from liquidity considerations, transaction costs and other market microstructure effects.

¹⁶ Other important theoretical work on takeovers is, among others, Grossman and Hart (1980), Shleifer and Vishny (1986) and Stultz (1988).

¹⁷ In a debt buyback, the quality of the outstanding debt increases with the extent of the buyback. This has "reverse" Coasian dynamics: the fact that the borrower may buy back more debt in the future raises the cost of the buyback today. This may raise the cost of the buyback to a level at which efficient buybacks do not occur.

trade takes place prior to the resolution of uncertainty. This paper differs in that uncertainty is resolved continuously, and simultaneously with trade.¹⁸ In addition, Section 7 of this paper shows how our model can also be applied to the standard durable goods monopoly problem, and shows that our results characterize the equilibrium for the case of a general cost structure for the monopolist.

Some Empirical Evidence. The first issue that one needs to address is whether large shareholders play important role in the contemporary global economy. The answer is a resounding yes. Jensen and Warner (1988) show that a few individuals typically own a large fraction of the corporate ownership structures in the United States. La Porta et al (1999) show that, globally, corporate ownership is, typically, even more concentrated than in the United States (with the exception of Britain which exhibits similar ownership patterns as the U.S.). For example, one family has a controlling interest in 5 out of 20 largest Swedish companies.¹⁹ Institutional analysis of large shareholder monitoring for the U.S. can be found in Black (1992) and for the U.K. in Black and Coffee (1994), among others.²⁰

How did ownership structure change over time? For American companies, the following is known: a) private companies going public have high concentration of managerial ownership, which gradually, but significantly, declines over time after the initial public offering (Mikkelson et al (1997)); and b) managerial ownership is, on average, higher now than it was in the 1930s.²¹

In this paper we have taken the point of view that, at least in the initial stages of a company's lifecycle, trading by the insiders is largely driven by their risk-sharing motivation rather than the asymmetric information between the insiders and the small outside investors. Is that a sensible assumption? Recently, Meulbroek (2000) performs an important empirical study aimed at determining motives for insider trading. She analyses the price impact of insider trades in the internet-related companies. She demonstrates that the insiders in these companies are much more frequently selling than buying their own stock. They are doing so, also, in comparison with other less volatile industries. In contrast to the results of Seyhun (1986) who demonstrates a negative price impact of insider sales on the stock prices in other less volatile industries, Meulbroek (2000) finds no such impact in her sample. She performs a series of tests which allow her to conclude

¹⁸ A very different model of monopoly is considered by Basak and Pavlova (2000). They consider a model of a firm that is a monopolist in the production of the consumption good. They consider the dynamic production policy that will maximize the firm's share price, when the firm (as in Basak (1996)) recognizes the impact of its production decisions on state prices, including the interest rate. They show that in the continuous time limit, the equilibrium share price of a time-consistent monopolist is zero. Importantly, they consider a representative agent economy, so that there is no trading of shares.

¹⁹ In many foreign countries (Belgium, Sweden, Korea, etc), there is a significant incidence of pyramidal ownership structure. Such structure allows large shareholders to have an absolute control of a company without owning 50% of the equity. Wolfenzon (1999) models the incentive tradeoffs for pyramidal ownership structure.

²⁰ La Porta et al (1999) argue that countries with weaker investor protection laws, such as Italy or Greece, tend to have more concentrated ownership than countries with relatively stronger investor protection laws such as Britain, *ceteris paribus*. They argue that large stakes can serve as a substitute for the legal protection of investors.

²¹ Holderness et al (1999) demonstrate that in the period from 1935-1995, mean top management stake in the major U.S. corporations increased from 14 to 21%.

that insiders in the internet companies are much more likely to trade for risk-diversification purposes than to capitalize on a private information. This conclusion is consistent with the conclusions of two prior studies (Elliot et al. (1984) and Givoly and Palmon (1985)) which demonstrate that managers do not appear to base their trades on material information about the company prior to important announcements (see Footnote 10 in Meulbroek (2000) for more details).

Regarding the relationship between ownership structure and firm performance, Mathiesen has collected a remarkable list of references.²² He classifies papers into five categories (although certainly not completely unambiguously), as primarily supporting one of the following hypotheses:²³

- I) *Corporate performance is an increasing function of managerial ownership* (25). Among them, Holthausen et al (1990) find a permanent rise in the stock price following a block purchase and a permanent drop following a block sale while Mikkelsen and Partch (1985) find a significant price drop the stock price at the initial announcement of the secondary offering; for publicly traded firms in the United States affirmative evidence can be found in Hertz and Smith (1993), among others; affirmative evidence from the listed French and German companies is presented in Franks et al (1995).
- II) *Corporate performance is a decreasing function of the managerial ownership* (13). For example, Johnson et al (1985) study negative impact of entrenchment, while Slovin and Sushka (1993) analyze cases when management-affiliated block-holders allow management to dilute share value away from other shareholders.
- III) *Corporate performance is a non-monotonous function of ownership concentration* (11). Some of the best-known examples in this category are: Morck et al (1988), Wruck (1989), Holderness et al (1999), etc. For example, Morck et al (1988) shows that Tobin's q, a measure of performance, initially increases as a function of ownership concentration, then decreases, and then, for sufficiently large values of the ownership stake, increases again.
- IV) *Reverse causality: managerial ownership is an increasing function of performance* (16). See Demsetz (1986), among others.
- V) *Mixed evidence and/or no relationship* (19). Sometimes, evidence is mixed: Barclay and Holderness (1989) find that 80% of the time a very significant market premium is paid for large blocks of shares; however, in 20% percent of the time large blocks trade *at a discount*. Other times, authors find no statistically significant relationship (Demsetz and Lehn (1985), Mikkelsen and Partch (1997), etc.).

In summary, although a significant number of collected references (41 out of 84) report a positive correlation between the corporate ownership and performance, the evidence for this relationship (and especially the causality) remains mixed. Our hope is that by

²² See his web site www.encycogov.com.

²³ In brackets is the number of papers which we find in each of the categories (we added, also, some references not on his list).

developing *dynamic* implications for holdings, we may be better able to sort out these issues empirically.

2. The Model

We consider a going concern firm in unit supply with a cumulative free cash flow process described by the diffusion

$$dD = \mu dt + \sigma dZ,$$

where Z is standard Brownian motion.²⁴ Shares of the firm trade in the market at price V , to be determined in equilibrium. The firm pays out all cash flows as dividends.²⁵ In addition to this firm there exists a riskless investment that pays a continuously compounded return of r , with perfectly elastic supply.

There is an agent in the model, interpreted as an owner-entrepreneur or a large shareholder in the firm with the ability to monitor the firm and affect decisions within the firm. In addition to this agent, there exists a continuum of competitive investors, with measure F .²⁶ Implicitly, it is assumed that infinitesimal shareholders cannot form coalitions or in any other way behave strategically.²⁷ All individuals in the economy have standard CARA utility so that on date t , they optimize

$$E_t \int_t^\infty e^{-R^i(t-t)} u^i(c_t) dt,$$

where $u^i(c) = -e^{-\gamma^i c}$, γ^i is the individual's coefficient of absolute risk aversion and R^i is the individual's rate of time preference. If $\gamma^i = 0$, we restrict $R^i = r$ to ensure the individual's problem has a solution. With a slight abuse of notation, we denote by γ^a the risk aversion of the agent, and by γ^i the risk aversion of investor i , and similarly for R^a and R^i .

The agent manages the firm and can trade shares. Let $\alpha(t) \in [0,1]$ denote the fraction of the firm held by the agent at time t . We restrict α to be right continuous, and interpret

$$\alpha(t-) \equiv \lim_{\tau \uparrow t} \alpha(\tau)$$

²⁴ That is, cash flows each period are normally distributed, and independent across periods. Independence is not crucial for our results, but simplifies the analysis.

²⁵ Holding fixed investment opportunities (so that any retained cash flows are invested in marketed securities), this assumption is for convenience only; dividend policy irrelevance holds in this setting. Note also that normality implies that dividends may be arbitrarily negative, implying that shareholders do not have limited liability. Normality is important for tractability, however, and while unlimited liability is unnatural, it does not play any significant role in the forces underlying our results.

²⁶ In order to simplify the analysis, we neglect the fact that the agent's trading opportunities depend, in principal, on the liquidity of the market which is, in turn, endogenous to her trading behavior due to microstructure effects (adverse selection, inventory cost etc). In Section 7 we discuss directional changes in our results which one would anticipate if such effects are taken into account.

²⁷ This is, clearly, a simplification of reality. Zwiebel (1995) shows that, when small block holders are allowed to co-exist with a large shareholder, there exists a threshold level of ownership by the large shareholder above which no coalition by the small block holders takes place. What we, essentially, assume is that such threshold is sufficiently small so that coalitions never occur.

as the shares held at the “start” of period t ; thus, $\alpha(t) - \alpha(t-)$ is the discrete number of shares purchased in period t . The agent has an initial endowment $\alpha(0-) = \alpha^-$. By market clearing, in equilibrium the investors’ holdings at time t are given by $1 - \alpha(t)$.

The mean dividend rate μ and the volatility σ of the firm may be altered by the actions of the agent. In particular, the mean dividend rate and the volatility are given by bounded functions $\hat{\mu}(e, t)$ and $\hat{\sigma}(e, t)$, where e corresponds to the agent’s instantaneous “effort” choice. Effort is potentially costly for the agent. This cost is represented by a reduction in consumption at rate $f(e, \alpha, t)$. Note that the cost may depend on α – it may be easier to monitor and implement changes if α is large, or there may be other benefits associated with control. We normalize f so that²⁸

$$0 = \min_e f(e, 0, t). \quad (1)$$

The investors consume and trade shares of the firm and the riskless security continuously and competitively. The agent may consume, make effort choices, and trade the riskless security continuously, but is restricted to trade shares of the firm on a finite set of dates $\mathbb{T}$.²⁹ The agent cannot commit to future effort choices (effort cannot be contracted on); thus, the agent’s effort choice must be incentive compatible. We will consider both the case when the agent can commit to a trading strategy, and the case when the agent is unable to commit to future trades and these decisions must be sequentially rational.

Though the investors trade competitively as price-takers, the investors are aware that the agent’s current trading decision has an impact on the agent’s current and future effort choices and trading decisions. We assume that the agent’s trading decisions are observable by the investors. The investors rationally take into account this information when determining their demands for shares.

Loosely speaking, the equilibrium for this economy can be described as follows. Given current shareholdings α at time t , the agent makes a trading decision and an effort choice. Based on the agent’s trading decision, investors competitively determine the share price V , taking into account the future trading decisions and effort choices of the agent. We will formalize this in the analysis below.

3. Preliminary Analysis

3.1. Competitive Share Price

Let $(\mu(t), \sigma(t))$ be arbitrary, bounded, deterministic mean and volatility components of the dividend process anticipated by the investors. In this section we determine the corresponding equilibrium share price process for the stock.

²⁸ Implicitly, we assume that the min exists, which is a necessary condition for existence of an optimal effort choice for the agent. Beyond that, this is a pure normalization and is without loss of generality.

²⁹ Technically, a finite set of trading dates allows us to use subgame perfection to identify a unique equilibrium. An economic justification is that trading restrictions, such as lockups, are frequently imposed on large shareholders so that their trading times are, typically, discrete. In addition, large venture capital investors usually have finite (and known) life span. After the partnership dissolves, partners may still stay invested in the company, but do not usually monitor their investment closely. This justifies a finite trading time horizon.

Recall that investors behave competitively and take the equilibrium stock price process as given. Given CARA utility, it is natural to conjecture a price process of the form

$$dV = (rV - \mu + \rho) dt, \quad (2)$$

or,

$$V(t) = \int_t^\infty e^{-r(\tau-t)} [\mu(\tau) - \rho(\tau)] d\tau,$$

where ρ represents the equilibrium risk premium.

Next we solve for the optimal demand for investor i . In the proof we introduce standard (non-binding) restrictions on the trading strategy to prevent “doubling strategies” and Ponzi schemes, necessitated by the continuous-time, infinite horizon framework. The result below confirms the intuition that the investor’s demand is increasing in the risk premium, and decreasing in volatility and risk aversion.

PROPOSITION 1. Given (μ, σ, V, ρ) with ρ/σ^2 bounded, investor i has optimal demand (subject to non-binding trading restrictions) given by

$$\alpha^i = \rho / (\gamma^i r \sigma^2),$$

with α^i arbitrary if $\rho = \sigma^2 = 0$. The investor’s expected payoff on date t is given by $r^{-1} u^i(r [W + k^i(t)] + (R^i - r)/(\gamma^i r))$, where the certainty equivalent $k^i(t)$ is

$$k^i(t) = \int_t^\infty e^{-r(s-t)} \left[\frac{1}{2} \alpha^{i2} \gamma^i r \sigma^2 \right] ds = \int_t^\infty e^{-r(s-t)} \frac{\rho^2}{2\gamma^i r \sigma^2} ds.$$

PROOF: Let $|\rho/\sigma^2| \leq M$. To eliminate doubling strategies, we bound the investor’s position $|\alpha^i| \leq M/(\gamma^i r)$. Since σ is bounded by assumption, this implies that $|\alpha^i \sigma|$ and ρ^2/σ^2 are bounded as well.

Given a trading strategy α and a consumption profile c , the riskless bond holdings of the investor evolve according to

$$dB = (rB + \alpha\mu - c) dt - V d\alpha + \alpha\sigma dZ.$$

Thus, the investor’s wealth $W = B + \alpha V$ evolves according to

$$dW = (rW + \alpha\rho - c) dt + \alpha\sigma dZ.$$

Consider the value function,

$$J(W, t) = r^{-1} u^i(r [W + k^i(t)] + (R^i - r)/(\gamma^i r)).$$

The Bellman equation is

$$\max_{\alpha, c} J_W [rW + \alpha\rho - c] + J_t - R^i J + \frac{1}{2} \alpha^2 \sigma^2 J_{WW} + u^i(c_t) = 0.$$

Maximizing over c yields,

$$J_W = u^{i'}(c_t).$$

Since $J_W = u''(r [W + k^i(t)] + (R^i - r)/(\gamma^i r))$,

$$c_t = r [W + k^i(t)] + (R^i - r)/(\gamma^i r).$$

Maximizing over α , and using the fact that $J_W = -\gamma^i r J$, yields as the desired equation for equilibrium demand

$$\rho = -\alpha \sigma^2 J_{WW}/J_W = \alpha \gamma^i r \sigma^2.$$

Finally, we need to check that the Bellman equation holds. Using the above, together with $J_t = -\gamma^i r J k^{i'}$ and $u^i(c_t) = r J$, it can be rewritten as

$$k^{i'} = r k^i - \frac{1}{2} \alpha^2 \sigma^2 r \gamma^i.$$

This has the solution,

$$k^i(t) = \int_t^\infty e^{-r(s-t)} \left[\frac{1}{2} \alpha^2 \sigma^2 r \gamma^i \right] ds + k_0^i e^{rt},$$

where k_0^i is an arbitrary constant of integration. This yields the wealth process,

$$dW = \left(\frac{1}{2} \alpha^2 \sigma^2 r \gamma^i - (R^i - r)/(\gamma^i r) - k^{i'} \right) dt + \alpha \sigma dZ.$$

Since $\alpha^2 \sigma^2$ is bounded, we have

$$E W_T = O(T) - k_0^i e^{rT}.$$

To avoid Ponzi schemes we restrict the investor to strategies for which the present value of future debt goes to zero; this requires $k_0^i \leq 0$. Investor optimality then implies $k_0^i = 0$, yielding the desired result.

Verification of the Bellman equation is straightforward if standard restrictions on the budget set are imposed so that $E[\exp(-R^i T) J(W_T^{\hat{\alpha}, \hat{c}}, T)] \rightarrow 0$. ♦

Using this result, the equilibrium risk premium for the stock follows from the market clearing condition. Let α be the current shareholdings of the agent. Then the aggregate demand of the investors must satisfy

$$1 - \alpha = \int_i \alpha^i dF(i) = \rho / (r \sigma^2) \int_i 1/\gamma^i dF(i) = \rho / (\gamma^J r \sigma^2),$$

where

$$\gamma^J \equiv \left[\int_i 1/\gamma^i dF(i) \right]^{-1}.$$

Thus we have the equilibrium risk premium,

$$\rho = (1 - \alpha) \gamma^J r \sigma^2.$$

Determining the equilibrium share price requires the risk premium process for all future dates. From the above, this follows from the equilibrium supply of shares based on the agent's trading strategy. Given a deterministic equilibrium path $\alpha(t)$ for the agent, we have the following equilibrium share price:

PROPOSITION 2. Given deterministic (α, μ, σ) , the equilibrium share price is given by^{30,31}

$$V(t | \alpha, \mu, \sigma) = \int_t^\infty e^{-r(\tau-t)} [\mu(\tau) - \rho^I(\tau)] d\tau,$$

where $\rho^I(\tau) = (1 - \alpha(\tau)) \gamma^I r \sigma^2(\tau)$.

PROOF: Follows from the above. Note that ρ/σ^2 is bounded, as required. ♦

The above pricing formula is natural: the share price is the present value of expected future dividends, adjusted by a risk premium that depends upon the quantity of risk held by investors. Note however, that both expected dividends and risk premia depend upon the agent's anticipated trading strategy, which must be determined in equilibrium.

3.2. Agent's Consumption and Effort Choice

In this section we evaluate the agent's optimal consumption and effort choices given a fixed deterministic trading profile $\alpha(t)$. Let $\mathbb{T} = \{t_1 = 0, t_2, \dots, t_N = T\}$ be the finite set of trading dates. In this case, $\alpha(t) = \alpha(t_i)$ for all $t \in [t_i, t_{i+1})$, where we define $t_{N+1} = \infty$. We also assume that the agent anticipates a share price process $V(t)$, which gives the share price on date t .

We now consider the agent's optimal effort and consumption decisions. This requires solving for the agent's optimal holdings of the riskless asset, which can be adjusted continuously. In particular, the agent solves

$$\begin{aligned} J(B_t, t | \alpha, V) = & \max_{c, e} E \int_t^\infty e^{-R^a(\tau-t)} u(c_\tau) d\tau \\ \text{s.t.} \quad & dB = (rB + \alpha\mu - f - c)dt - Vd\alpha + \alpha\sigma dZ. \end{aligned}$$

In solving this problem, it is useful to define the following

$$z(\alpha, t) \equiv \max_e \alpha \hat{\mu}(e, t) - f(e, \alpha, t) - \frac{1}{2} \alpha^2 \gamma^I r \hat{\sigma}^2(e, t). \quad (3)$$

This represents the optimal, risk-adjusted, dividend process net of effort costs that the agent can achieve. Let $e(\alpha, t)$ be the corresponding effort choice. We make the following technical assumptions, which we maintain throughout.

³⁰ Note that the share price process is deterministic. This is because the firm does not retain earnings, and future cash flows are independent of the past (there is no learning from past cash flows about the future). Relaxing either of these assumptions would lead to a stochastic share price. For example, if the firm had accumulated retained earnings $A(t)$, the share price would be $S(t) = V(t) + A(t)$ and would be stochastic if $A(t)$ depends on past cash flows. This would complicate notation, *but otherwise not change the model* – investors and the agent would simply borrow against $A(t)$. A similar analysis would apply if the agent's effort did not generate cash at each date t but instead improved the “terminal value” of the firm.

³¹ Note that the analysis is unchanged if the investors are restricted to trading shares only on dates $\mathbb{T}$, since in equilibrium, investors only trade when the agent trades.

ASSUMPTION A. For all (α, t) , $e(\alpha, t)$ is unique. In addition, $z(\cdot, t)$ is bounded and absolutely continuous in α .

We have the following characterization of the agent's effort and consumption:

PROPOSITION 3. Given deterministic (α, V) , the agent's optimal effort is given by $e = e(\alpha(t), t)$. The value function is given by

$$J(B_t, t | \alpha, V) = \frac{1}{r} u \left(r \left[B_t + k(t | \alpha, V) \right] + \frac{1}{\gamma^a r} (R^a - r) \right),$$

where the certainty equivalent $k(t)$ is given by³²

$$k(t | \alpha, V) = \int_{[t, \infty]} e^{-r(\tau-t)} \left[z(\alpha(\tau), \tau) d\tau - V(\tau) d\alpha(\tau) \right].$$

PROOF: Consider the interval $[t_i, t_{i+1})$. On this interval, shareholdings are held constant; i.e., $\alpha(t) = \alpha(t_i) \equiv \alpha_i$. The Bellman equation is

$$\max_{c, e} J_B \left[rB + \alpha_i \mu - f - c \right] + J_t - rJ + \frac{1}{2} \alpha_i^2 \sigma^2 J_{BB} + u(c_t) = 0.$$

Maximizing over c yields,

$$J_B = u'(c)$$

Using the shorthand $k(t)$ for $k(t | \alpha, V)$, $J_B = u'(r [B + k(t)])$, and thus we have the desired expression for c ,

$$c_t = r [B + k(t)].$$

Using $J_B = -r \gamma^a J$, $u(c_t) = r J$, and $J_t = -r \gamma^a J K'$, we can rewrite the Bellman equation as

$$-r \gamma^a [\alpha_i \mu - f - rk] - r \gamma k' + \frac{1}{2} \alpha_i^2 \sigma^2 r^2 \gamma^{a^2} = 0.$$

Maximizing over e yields $e = e(\alpha(t), t)$ and thus

$$k' = r k - z(\alpha(t), t).$$

Thus, we have

$$k(t) = \int_t^{t_{i+1}} e^{-r(s-t)} z(\alpha(s), s) ds + e^{-r(t_{i+1}-t)} k(t_{i+1}).$$

For the case $i = N$, $t_{N+1} = \infty$ and the last term becomes $k_0 e^{rt}$. As in the proof of **PROPOSITION 1**, the transversality condition and borrowing constraints imply $k_0 = 0$, and a similar verification argument holds.

³² We adopt set notation for the limits of integration to avoid ambiguity given discontinuities in α . Thus, $\int_{(t, T]} d\alpha(\tau) = \alpha(T) - \alpha(t)$ and $\int_{[t, T)} d\alpha(\tau) = \alpha(T-) - \alpha(t-)$, where $\alpha(t-) = \lim_{\tau \uparrow t} \alpha(\tau)$.

To incorporate the trading gains, we proceed by induction. Suppose the value function holds at trading date t_{i+1} . Now consider date $t = t_i$. At this date the agent also receives additional income of $-(\alpha_i - \alpha_{i-1}) V(t_i)$ from trading shares of the firm. This is equivalent to $\int_{[t_i, t_{i+1})} -V(s) d\alpha(s)$. Thus, we have

$$k(t_i) = \int_{[t_i, t_{i+1})} e^{-r(s-t_i)} \left[z(\alpha(s), s) ds - V(s) d\alpha(s) \right] + e^{-r(t_{i+1}-t_i)} k(t_{i+1}),$$

which gives the desired form. ♦

3.3. A Reduced Form

From the previous section, given a trading process $\alpha(t)$ for the agent, optimal effort on date t is given by $e(\alpha(t), t)$. Knowing this, investors can anticipate μ and σ , given α . Thus we define, with a slight abuse of notation,

$$\mu(\alpha, t) \equiv \hat{\mu}(e(\alpha, t), t),$$

$$\sigma^2(\alpha, t) \equiv \hat{\sigma}^2(e(\alpha, t), t),$$

as the dividend rate and volatility the agent will choose given α and t . We also introduce

$$b(\alpha, t) \equiv -f_\alpha(e, \alpha, t) \Big|_{e=e(\alpha, t)}$$

as the marginal control benefit the agent earns from holding shares. Clearly, given the primitives $(\hat{\mu}, \hat{\sigma}, f)$, we can derive the functions (μ, σ, b) . In fact, in many circumstances, it is convenient to take (μ, σ, b) as a reduced form for the underlying moral hazard problem. The following result verifies the equivalence of this approach, and shows how z may be computed in terms of (μ, σ, b) .

PROPOSITION 4. Given any (μ, σ, b) , there exists a consistent $(\hat{\mu}, \hat{\sigma}, f)$. Given any $(\hat{\mu}, \hat{\sigma}, f)$ consistent with (μ, σ, b) ,

$$z(\alpha, t) = \int_0^\alpha z_\alpha(\hat{\alpha}, t) d\hat{\alpha},$$

where

$$z_\alpha(\alpha, t) = \mu(\alpha, t) - \alpha \gamma^a r \sigma^2(\alpha, t) + b(\alpha, t).$$

PROOF: Given (μ, σ, b) , let $e \in \mathfrak{R}^2$ and $\hat{\mu}(e, t) = e_1$ and $\hat{\sigma}^2(e, t) = e_2$.

Also define

$$\begin{aligned} f(e, \alpha, t) &= \alpha e_1 - \frac{1}{2} \alpha^2 \gamma^a r e_2 + (e_1 - \mu(\alpha, t))^2 + (e_2 - \sigma^2(\alpha, t))^2 \\ &\quad - \int_0^\alpha \left[\mu(\hat{\alpha}, t) - \hat{\alpha} \gamma^a r \sigma^2(\hat{\alpha}, t) + b(\hat{\alpha}, t) \right] d\hat{\alpha}. \end{aligned}$$

Then

$$\begin{aligned} & \max_e \alpha \hat{\mu}(e, t) - f(e, \alpha, t) - \frac{1}{2} \alpha^2 \gamma^r r \hat{\sigma}^2(e, t) \\ &= \max_e - (e_1 - \mu(\alpha, t))^2 - (e_2 - \sigma^2(\alpha, t))^2 + \int_0^\alpha [\mu(\hat{\alpha}, t) - \hat{\alpha} \gamma^r r \sigma^2(\hat{\alpha}, t) + b(\hat{\alpha}, t)] d\hat{\alpha}. \end{aligned}$$

This is clearly solved with $e(\alpha, t) = (\mu(\alpha, t), \sigma^2(\alpha, t))$. Thus, μ and σ^2 are correct. Also,

$$\begin{aligned} -f_\alpha(e, \alpha, t) &= -e_1 + \alpha \gamma^r r e_2 + \mu(\alpha, t) - \alpha \gamma^r r \sigma^2(\alpha, t) \\ &\quad + b(\alpha, t) + 2(e_1 - \mu(\alpha, t))\mu_\alpha(\alpha, t) + 2(e_2 - \sigma^2(\alpha, t))\sigma_\alpha^2(\alpha, t) \\ &= b(\alpha, t). \end{aligned}$$

for $e = e(\alpha, t)$. Thus, (μ, σ, b) is consistent with $(\hat{\mu}, \hat{\sigma}, f)$.

Finally, for any $(\hat{\mu}, \hat{\sigma}, f)$ with corresponding (μ, σ, b) , $z_\alpha(\alpha, t)$ follows from the definition of z and the envelope theorem. The equation for z then follows by integration, given the normalization of f in (1) and absolute continuity (**ASSUMPTION A**). ♦

Next define,

$$v(\alpha, t) \equiv \mu(\alpha, t) - (1 - \alpha) \gamma^r r \sigma^2(\alpha, t),$$

which represents the risk-adjusted dividend rate anticipated by investors given shareholdings α for the agent. We assume that v is increasing in α , or equivalently, that the marginal value of the dividend stream to investors is decreasing in their own holdings $1 - \alpha$.

ASSUMPTION B. For all α and t , $v_\alpha(\alpha, t) > 0$.

This assumption is satisfied in the natural case that μ is increasing and σ^2 is decreasing (or constant) in α .

Finally, we conclude this section by characterizing the equilibrium share price and the agent's utility *given* a trading policy α :

PROPOSITION 5. Given α , the equilibrium share price is given by

$$V(t) = \int_t^\infty e^{-r(\tau-t)} v(\alpha(\tau), \tau) d\tau,$$

and the agent's equilibrium certainty equivalent payoff can be written as

$$k(t) = \int_t^\infty e^{-r(\tau-t)} [z(\alpha(\tau), \tau) + (\alpha(t-) - \alpha(\tau)) v(\alpha(\tau), \tau)] d\tau, \quad (4)$$

where $\alpha(t-)$ is the agent's shareholdings at the start of period t .

PROOF: The expression for V follows immediately from **PROPOSITION 2** and the definition of v . Using this expression for V in **PROPOSITION 3** yields,

$$\begin{aligned}
k(t) &= \int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau), \tau) d\tau - \int_{[t, \infty]} e^{-r(\tau-t)} V(\tau) d\alpha(\tau) \\
&= \int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau), \tau) d\tau - \int_{[t, \infty]} e^{-r(s-t)} \int_s^\infty e^{-r(\tau-s)} v(\alpha(\tau), \tau) d\tau d\alpha(s) \quad (5) \\
&= \int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau), \tau) d\tau - \int_t^\infty e^{-r(\tau-t)} \left(\int_{[t, \tau]} d\alpha(s) \right) v(\alpha(\tau), \tau) d\tau \\
&= \int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau), \tau) d\tau + \int_t^\infty e^{-r(\tau-t)} (\alpha(t-) - \alpha(\tau)) v(\alpha(\tau), \tau) d\tau,
\end{aligned}$$

as required. ♦

3.4. Benchmark Trading Strategies

In this section, we use the results thus far to solve for trading strategies in a number of important benchmark cases. First, we consider the optimal trading strategy if the agent can commit to a trading policy in advance. Second, we consider the optimal trading policy that would be designed by a social planner. Third, we derive the price-taking strategy, and show conditions for which a Walrasian equilibrium exists. These results are useful in demonstrating the relationship between our model and a number of others in the literature. Moreover, they will play a critical role in understanding the optimal time-consistent trading policy that we will analyze in Section 4.

Suppose the agent can commit in advance to a trading strategy. That is suppose that at time t , the agent announces a trading process $\alpha(\tau)$, $\tau \geq t$, and *cannot revise* this trading strategy in the future. Then we can use the result of **PROPOSITION 5** to determine the utility of the agent for each choice of α . The one that maximizes the agent's utility is the optimal **commitment strategy**:³³

PROPOSITION 6. If the agent commits to strategy α from t to T , the certainty equivalent payoff is given by (4). Thus, the optimal trading strategy for the agent to commit to in period t is given by

$$\alpha^c(\tau | \alpha(t-)) = \operatorname{argmax}_\alpha z(\alpha, \tau) + (\alpha(t-) - \alpha) v(\alpha, \tau),$$

for $\tau \in [t, T]$. The corresponding first order condition is

$$z_\alpha - v = -(\alpha(t-) - \alpha^c) v_\alpha.$$

Note also that $\alpha^c(\tau | \alpha(t-))$ is increasing in $\alpha(t-)$.

³³ This result generalizes (to a dynamic setting) the result of Admati, et al. (1994) in the case of a single risk asset (see their equation (5)).

PROOF: Follows from **PROPOSITION 5**. For the comparative static on α^c , note that

$$\max_{\alpha} z(\alpha, \tau) + (\alpha(t-) - \alpha)v(\alpha, \tau)$$

is convex in $\alpha(t-)$ (as a max of linear functions), with subgradient v . This implies that $v(\alpha^c, \tau)$ is increasing in $\alpha(t-)$, and therefore, by **ASSUMPTION B**, α^c must also be increasing. ♦

Note that if both investors and the agent are risk neutral ($\gamma^I = \gamma^A = 0$) and there are no control benefits ($b = 0$), then $z_{\alpha} = v = \mu$ and there will be no trade in this environment; that is, $\alpha^c = \alpha(t-)$. Thus, risk aversion and/or control benefits are necessary to create the wedge in agent's and investors' valuation that will induce trade. On the other hand, the difference in marginal valuations of the dividend streams, $z_{\alpha} - v$, is a measure of the potential gains from trade. If $z_{\alpha} - v \neq 0$, then the agent will always trade some amount ($\alpha^c \neq \alpha(t-)$).

Another useful benchmark is the case when the agent acts as a price-taker, ignoring the impact of α on the share price V . Intuitively, at the price-taking equilibrium, the agent's and the investors' marginal value for a share should be equated so that there is no motive for trade. That is, $\alpha^P(t)$ should satisfy,

$$z_{\alpha}(\alpha^P(t), t) - v(\alpha^P(t), t) = 0. \quad (6)$$

To be sure that α^P is well-defined, we introduce the following assumption, which states that z_{α} and v cross at a single point:

ASSUMPTION C. For all t , there is a unique $\alpha^P(t)$ satisfying (6). Further, for all $\alpha > (<) \alpha^P(t)$, $z_{\alpha}(\alpha, t) - v(\alpha, t) < (>) 0$.

The fact that $\alpha^P(t)$ equates the marginal valuations of both the agent and the investors suggests that it should be the outcome of a competitive, Walrasian equilibrium. The next result establishes that this is the case *if a Walrasian equilibrium exists*:

PROPOSITION 7. A strategy $\alpha^P(t)$ satisfying (6) is a price-taking Walrasian equilibrium if and only if

$$\alpha^P(t) = \operatorname{argmax}_{\alpha} z(\alpha, t) - \alpha v(\alpha^P(t), t), \quad (7)$$

for almost all t . This has the first order condition (6), which is sufficient if $z_{\alpha\alpha} \leq 0$. If $z_{\alpha\alpha} > 0$, then no Walrasian equilibrium exists.

PROOF: Suppose investors anticipate that the agent will trade according to α^P . Then the equilibrium share price is given by,

$$V^P(t) = \int_t^{\infty} e^{-r(s-t)} v(\alpha^P(s), s) ds.$$

Taking the share price as given, from **PROPOSITION 5** the agent solves:

$$\max_{\alpha} \int_t^{\infty} e^{-r(\tau-t)} \left[z(\alpha(\tau), \tau) + (\alpha(t-) - \alpha(\tau)) v(\alpha^p(\tau), \tau) \right] d\tau.$$

Thus, the equilibrium condition $\alpha(\tau) = \alpha^p(\tau)$ is equivalent to (7). Then (6) is clearly the first order condition, and the second order conditions are determined by the sign of $z_{\alpha\alpha}$.

◆

Thus, α^p is the price-taking equilibrium trading strategy when such a strategy exists. Note that such an equilibrium may not exist in our setting, however. Unlike a standard exchange economy where non-existence of equilibrium is “pathological,” here it arises naturally as a result of the moral hazard problem. Indeed, $z_{\alpha\alpha}$ is positive – making a Walrasian equilibrium impossible – if μ_{α} is sufficiently positive; that is, if the agent has a large enough impact on the dividend stream.³⁴ The fact that the Walrasian equilibrium may fail to exist reveals the untenable nature of the price-taking assumption in the case of a large shareholder. Nonetheless, α^p serves as a useful benchmark in our analysis of the equilibrium when the large shareholder is strategic.

For example, comparing the commitment strategy α^c with α^p , it is straightforward to show that³⁵

$$\alpha^c(t) \in (\alpha(t-), \alpha^p(t)],$$

that is, that α^c is between the initial holdings and α^p . This corresponds to the fact that the agent, as a monopolist in choosing his holdings, refrains from trading all the way to the competitive solution in order to receive a better price.

Note that we can rewrite (6) to define $\alpha^p(t)$ as the solution to

$$\alpha^p(t) = \frac{\gamma^I}{\gamma^*} + \frac{b(\alpha^p(t), t)}{\gamma^* r \sigma^2(\alpha^p(t), t)}, \quad (8)$$

where $\gamma^* = \gamma^I + \gamma^J$. Thus, if there are no control benefits and $b = 0$, then $\alpha^p = \gamma^I / \gamma^*$, the optimal risk sharing solution. Because α^p is independent of μ , this solution does not account for the agent’s incentives, and so is generally inefficient. To see this, we derive next the socially optimal trading strategy.³⁶

PROPOSITION 8. The socially-optimal trading strategy for the agent is given by

$$\alpha^s(t) = \operatorname{argmax}_{\alpha} z(\alpha, t) + (1 - \alpha) v(\alpha, t) + \frac{1}{2} (1 - \alpha)^2 \gamma^I r \sigma^2,$$

for almost all t . This has the first order condition

³⁴ Intuitively, given a fixed price and z convex, the agent will choose either $\alpha = 1$ (if the price is low) or $\alpha = 0$ (if the price is high). But since the market price is low if $\alpha = 0$ and high if $\alpha = 1$, this is inconsistent with an equilibrium.

³⁵ Note that we have written this heuristically in that $\alpha(t-)$ may be larger than α^p ; the interpretation should be clear.

³⁶ This is consistent with the static results of Admati, et al. (1994, equations 15-16).

$$\begin{aligned} z_\alpha - v &= - (1 - \alpha^s) v_\alpha + (1 - \alpha^s) \gamma^I r \sigma^2 - \frac{1}{2} (1 - \alpha^s)^2 \gamma^I r \sigma_\alpha^2 \\ &= - (1 - \alpha^s) \mu_\alpha + \frac{1}{2} (1 - \alpha^s)^2 \gamma^I r \sigma_\alpha^2. \end{aligned}$$

PROOF: Note that **PROPOSITION 1** implies that the investors' aggregate certainty equivalent is given by $(1 - \alpha(t-)) V(t) + \int_t^T k^I(t) dF(i)$. From **PROPOSITION 2**, this is equal to

$$(1 - \alpha(t-)) V(t) + \int_t^T e^{-r(s-t)} \frac{1}{2} (1 - \alpha(s))^2 \gamma^I r \sigma^2 ds.$$

The result then follows on substituting equilibrium $V(t)$ and adding the agent's certainty equivalent from **PROPOSITION 5**. ♦

The socially optimal trading strategy maximizes the total certainty equivalent of both the agent and the investors. Any outcome on the Pareto Frontier can therefore be supported using this strategy together with appropriate transfer payments between the agent and the investors. Thus, this solution can be viewed as the outcome that would occur if the agent and the investors bargain over a contract that specifies a trading strategy. For example, this coincides with the outcome of a standard principal-agent problem in which the agent is compensated with both cash and shares of the firm.³⁷

Note that if there is no moral hazard problem ($\mu_\alpha = \sigma_\alpha^2 = 0$), then $\alpha^s = \alpha^p$. That is, the price-taking solution is efficient, as we know it must be by the First Welfare Theorem. With moral hazard, this is no longer the case. In the usual case in which the large shareholder enhances the dividend stream ($\mu_\alpha > 0$ and $\sigma_\alpha^2 \leq 0$), then $\alpha^s > \alpha^p$.

Comparing α^s with α^c , we note that the two coincide if $\alpha(t-) = 1$ and investors are risk neutral ($\gamma^I = 0$), since in this case investor surplus is zero.³⁸ If $\alpha(t-) < 1$, then $\alpha^c < \alpha^s$, and the agent will sell too much since he does not care about the price impact on the investors' initial holdings. Alternatively, if investors are risk averse ($\gamma^I > 0$) and $\sigma_\alpha^2 \leq 0$, then $\alpha^c > \alpha^s$, and the agent sells too little since he does not care about the surplus the investors receive from their infra-marginal holdings.

4. Optimal Trading Without Commitment

In **PROPOSITION 6** of the previous section, we solved for an optimal trading strategy assuming the agent could commit ex-ante to future trades. In that case, current share price depends on all future trades the agent will make. Here, we no longer allow the agent to commit to future trades. Thus, the agent's trading strategy must be time

³⁷ For example, in the special case of $\gamma^I = b = 0$ with μ independent of t and constant volatility σ , then α^s is the solution to

$$(1 - \alpha) \mu'(\alpha) / \alpha = \gamma^I r \sigma^2,$$

which coincides with the optimal principal-agent contract of Holmstrom and Milgrom (1987, Theorem 7). (In their case, $e \equiv \mu$ and so $\alpha = f'(\mu)$ and $\mu'(\alpha) = 1/f''(\mu)$.) Holmstrom and Milgrom show that this linear contract is optimal among all contracts.

³⁸ This generalizes similar static results of Kihlstrom and Matthews (1990) and Magill and Quinzii (2000). These authors also define an equilibrium concept in which the agent is a price-taker solely with respect to the risk premium. In this case, $\alpha^c = \alpha^s$ even with $\gamma^I > 0$.

consistent. The previous results suggest that the commitment policy α^c is not time consistent. To see the intuition for this, recall that with commitment, starting from any initial shareholdings the agent will trade towards the price-taking solution α^p . But this same argument implies that after the initial trade, the agent has an incentive to trade again, moving from his current position towards α^p .

For example, suppose the agent initially holds the entire firm ($\alpha^- = 1$). Then in the commitment case, the agent refrains from selling all the way to the price-taking solution ($\alpha^c > \alpha^p$) due to the impact his trade has on the value of his shares. However, once the trade is consummated he is willing to sell again since he now only suffers the price impact on the smaller number of shares $\alpha^c < 1$. He does not internalize the negative externality imposed on the shareholders who bought in the first round.

To solve for the equilibrium without commitment, note that the value of the shares at any time t must depend on investors' expectations of the agent's future trading decisions. Thus, investors must anticipate the agent's ex-post incentives to trade.

At any time t , the set of future possibilities depends only on the current shareholdings α and t . Suppose then, that the share price can be written as $V(\alpha, t)$.³⁹ Taking this as given, the agent optimally chooses a trading policy. Given the agent's optimal trading policy, we must then verify that V is correct in equilibrium.

From **PROPOSITION 3**, the agent's decision problem at time t , given $\alpha(t-)$, is to choose a policy $\alpha(\tau)$, $\tau \geq t$ to maximize the certainty equivalent⁴⁰

$$\int_{[t, \infty]} e^{-r(\tau-t)} [z(\alpha(\tau), \tau) d\tau - V(\alpha(\tau), \tau) d\alpha(\tau)].$$

Recall that the agent has the opportunity to trade only on the discrete dates $\mathbb{T} = \{t_1=0, t_2, \dots, t_N=T\}$. Implicitly, the agent therefore commits not to trade during the intervals (t_i, t_{i+1}) . We assume for simplicity that for $t \in [t_i, t_{i+1})$,⁴¹

$$b(\alpha, t) = b(\alpha, t_i), \quad \mu(\alpha, t) = \mu(\alpha, t_i), \quad \sigma^2(\alpha, t) = \sigma^2(\alpha, t_i).$$

If we let $\alpha_j = \alpha(t_j)$ and $\delta_j = \frac{1}{r} \left(1 - e^{-r(t_{j+1}-t_j)}\right)$, the above objective can be written

$$Z(\alpha_{j-1}, t_j) \equiv \max \sum_{j \geq i} e^{-r(t_j-t_i)} [\delta_j z(\alpha_j, t_j) + V(\alpha_j, t_j)(\alpha_{j-1} - \alpha_j)]. \quad (9)$$

This leads to the following first-order condition for α_j :

$$\delta_j z(\alpha_j, t_j) + e^{-r(t_{j+1}-t_j)} V(\alpha_{j+1}, t_{j+1}) - V(\alpha_j, t_j) + V_{\alpha}(\alpha_j, t_j) (\alpha_{j-1} - \alpha_j) = 0.$$

Along the optimal path, consistency of V implies (from **PROPOSITION 5**) that

³⁹ This is not restrictive if there is a unique equilibrium given (α, t) . Otherwise, this represents a “Markovian” assumption. See the further discussion in footnote 43.

⁴⁰ We could alternatively formulate this as a standard dynamic programming problem. Here, the “no commitment” constraint is embedded in the specification of V as a function of the current shareholdings only (with commitment, today's share price will also depend on today's commitment to future holdings).

⁴¹ Recall that we define $t_{N+1} = \infty$.

$$V(\alpha_j, t_j) = \delta_j v(\alpha_j, t_j) + e^{-r(t_{j+1}-t_j)} V(\alpha_{j+1}, t_{j+1}).$$

Combining with the above yields,

$$z_\alpha(\alpha_j, t_j) - v(\alpha_j, t_j) = V_\alpha(\alpha_j, t_j) (\alpha_j - \alpha_{j-1}) / \delta_j. \quad (10)$$

Notice that since trading ceases in period T , $V(\alpha_T, T) = v(\alpha_T, T)/r$. Thus, α_T is the commitment solution, given α_{N-1} .⁴² Given $V(\cdot, t_j)$, $j \geq i$, the solution to (9) defines optimal holdings $\alpha_j(\alpha_{j-1})$ as a function of the prior holdings, which is characterized by equation (10) if the first order conditions are necessary and sufficient. Since investors can anticipate this trade, $V(\cdot, t_{j-1})$ is given by

$$V(\alpha_{j-1}, t_{j-1}) = \delta_{j-1} v(\alpha_{j-1}, t_{j-1}) + e^{-r(t_j-t_{j-1})} V(\alpha_j(\alpha_{j-1}), t_j). \quad (11)$$

Thus, starting from $V(\cdot, T)$, equations (10) and (11) can be used recursively to construct the equilibrium.⁴³ Note that this coincides with the commitment case when there is only a single opportunity to trade, since then (10) reduces to $z_\alpha - v = v_\alpha (\alpha - \alpha^-)$.

In order to proceed with this characterization, however, we need conditions that verify the necessity of the first order condition (10) at the solution. In particular, we need to guarantee that the solution to (9) is interior. The following result establishes this:

PROPOSITION 9. In a time consistent equilibrium, α_j increases with α_{j-1} and $V_\alpha(\alpha_j, t_j) \geq \delta_j v_\alpha(\alpha_j, t_j) > 0$. This implies that the solution to (9) is interior, condition (10) holds, and

$$\begin{aligned} \alpha_{j-1} > \alpha_j^p &\Rightarrow \alpha_{j-1} > \alpha_j > \alpha_j^p, \\ \alpha_{j-1} = \alpha_j^p &\Rightarrow \alpha_{j-1} = \alpha_j = \alpha_j^p, \\ \alpha_{j-1} < \alpha_j^p &\Rightarrow \alpha_{j-1} < \alpha_j < \alpha_j^p. \end{aligned}$$

In addition, the agent's certainty equivalent payoff Z on date j is convex in the initial holdings α_{j-1} , with subgradient $V(\alpha_j(\alpha_{j-1}), t_j)$.

PROOF: First, note that the objective for (9) is linear in α_{j-1} . Thus, the agent's certainty equivalent payoff (9) is convex in α_{j-1} . This implies that the subgradient, $V(\alpha_j(\alpha_{j-1}), t_j)$ is

⁴² Recall that $\delta_T = 1/r$, so that (10) is equivalent to the first order condition in **PROPOSITION 6**.

⁴³ For each date t , if the solution to (9) is unique then the equilibrium constructed here is the unique subgame perfect equilibrium. This will occur generically in the parameters of the model; a sufficient condition given the conditions of Proposition 9 is that (10) have a unique solution. In the (non-generic) case of multiple solutions to (9), non-Markov equilibria are possible in which the agent's future trading strategy depends on the entire history of trades, rather than just the current holdings. Note also that a finite trading horizon prevents "Folk Theorem" multiplicity (see also Ausubel and Deneckere (1989)).

increasing in α_{j-1} , for all j . Then from (11) and **ASSUMPTION B**, we have $V_\alpha(\alpha_j, t_j) \geq \delta_j v_\alpha(\alpha_j, t_j) > 0$. Therefore, α_j is increasing in α_{j-1} .

Next, note that by **ASSUMPTION C**, for $\alpha_j \geq \alpha_{j-1} > \alpha^p_j$, the RHS of (10) is positive while the LHS is strictly negative. Thus it is optimal to decrease α_j . For $\alpha_j < \alpha^p_j < \alpha_{j-1}$, the RHS is negative and the LHS is strictly positive. Thus it is optimal to increase α_j . The similar arguments apply for the remaining cases. ♦

This result demonstrates that without commitment, in equilibrium the agent will continually trade toward the price-taking equilibrium α^p . Intuitively, the agent cannot resist the temptation to trade when the market's valuation for a share differs from the agent's private (marginal) valuation for a share. In the next section we use equations (10)-(11) to solve for an equilibrium explicitly in a special case.

4.1. Example: Linear Returns

In this subsection, we obtain a closed form solution for the system of Equations (10)-(11) for the special case in which

$$\mu(\alpha, t) = \mu_0(t) + \mu_1 \alpha, \quad \sigma^2(\alpha, t) = \sigma^2, \quad b(\alpha, t) = 0, \quad (12)$$

with $\mu_1 \geq 0$. We note that this is equivalent to the agent having a cost function associated with effort that is quadratic in the attained net increase in $\hat{\mu}$.⁴⁴ If $\mu_1 = 0$, this corresponds to the case of no moral hazard (as in Kihlstrom (1998)). We also assume the time increment between trading opportunities is constant,

$$\Delta = t_{i+1} - t_i \text{ for all } i.$$

Thus,

$$\delta_i = \delta = \frac{1}{r}(1 - e^{-r\Delta}) \text{ when } i \leq T-1, \text{ and } \delta_T = 1/r.$$

Recall that for this setting, $\alpha^p = \gamma^I/(\gamma^I + \gamma^J) = \gamma^I/\gamma^*$. In addition, the benchmark solutions of Section 3.4 are given by (starting from $\alpha^- = 1$),

$$\alpha^C = \frac{\mu_1 + 2\gamma^J r \sigma^2}{\mu_1 + 2\gamma^J r \sigma^2 + \gamma^J r \sigma^2},$$

$$\alpha^S = \frac{\mu_1 + \gamma^J r \sigma^2}{\mu_1 + \gamma^* r \sigma^2}.$$

Next we determine the no-commitment solution. For this case, equations (10)-(11) become:

$$\delta \gamma^* r \sigma^2 (\alpha^p - \alpha_t) = (\alpha_t - \alpha_{t-1}) V_\alpha(\alpha_t, t) \quad (13)$$

⁴⁴ Specifically, let $\hat{\sigma}^2(e, t) = \sigma^2$, $\hat{\mu}(e, t) = \mu_0(t) + e$, and $f(e, \alpha, t) = e^2 / (2 \mu_1)$. This is the model of Stoughton and Zechner (1998).

$$V(\alpha_{t-1}, t-1) = \delta(\mu_0(t-1) + \mu_1 \alpha_{t-1} - (1 - \alpha_{t-1}) \gamma^I r \sigma^2) + e^{-r\Delta} V(\alpha_t(\alpha_{t-1}), t) \quad (14)$$

In order to solve the system of Equations (13)-(14) one needs to solve simultaneously for the shape of V and for the optimal policy $\alpha(t)$. The following proposition provides an explicit characterization:

PROPOSITION 10. Let (μ, σ, b) be defined as in (12). Then

$$V(\alpha_t, t) = v_{1t} \alpha_t + v_{2t} \quad (15)$$

where the coefficients v_{it} satisfy the backward difference equation:

$$\begin{aligned} v_{1t-1} &= \delta(\mu_1 + \gamma^I r \sigma^2) + \exp(-r\Delta) \frac{v_{1t}^2}{v_{1t} + \delta \gamma^* r \sigma^2} \\ v_{2t-1} &= \delta(\mu_0(t-1) - \gamma^I r \sigma^2) + \exp(-r\Delta) \left[v_{2t} + \frac{v_{1t} \delta \gamma^I r \sigma^2}{v_{1t} + \delta \gamma^* r \sigma^2} \right] \end{aligned} \quad (16)$$

with $v_{1T} = (\mu_1 + \gamma^I r \sigma^2) / r$ and $v_{2T} = (\mu_0(T) - \gamma^I r \sigma^2) / r$. The optimal policy is the solution for the forward difference equation:

$$\alpha_t = \alpha^p + \frac{v_{1t}}{v_{1t} + \delta \gamma^* r \sigma^2} (\alpha_{t-1} - \alpha^p) \quad (17)$$

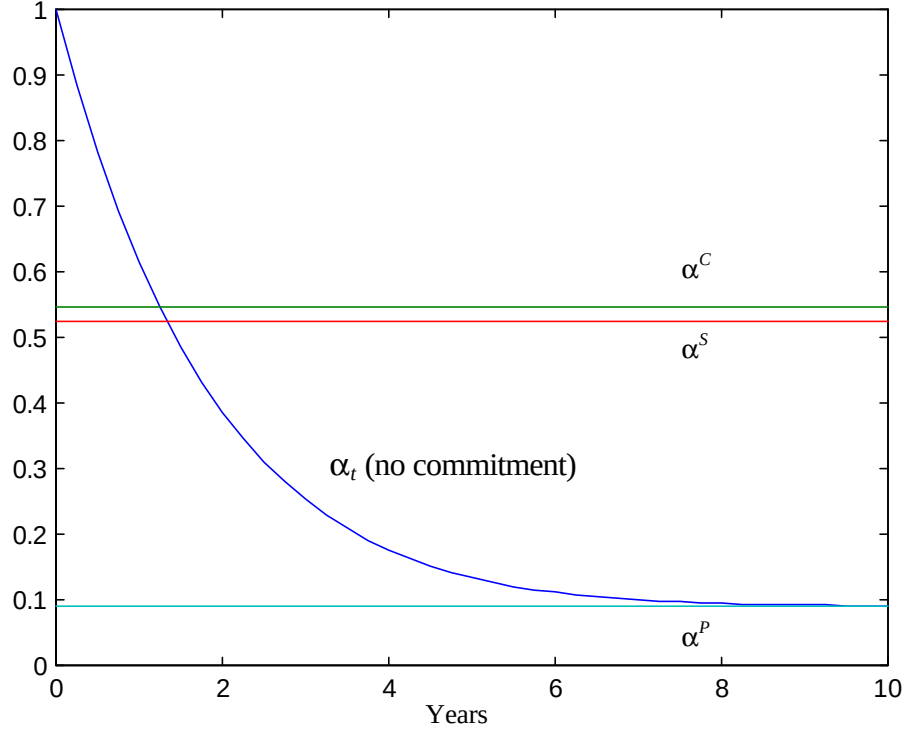
with the initial condition $\alpha_{-1} = \alpha^-$. Finally, α_T satisfies the above with $\delta = 1/r$.

PROOF: By induction. Since $V(\cdot, T) = v_{1T} \alpha_T + v_{2T}$, Equation (15) holds at T . Next, assume that Equation (15) is valid at time t . From Equations (13) and (15), Equation (17) follows for α_t . In turn, Equations (14) and (17) imply that Equations (15)-(16) hold for $t-1$. Thus, Equations (15)-(17) hold for all t . ♦

In the absence of moral hazard (i.e., when $\mu_1 = 0$), and restricted to three periods, (15)–(17) are equivalent to the no-commitment results of Kihlstrom (1998). In the presence of moral hazard, but restricted to two periods, (15)–(17) are equivalent to (63)–(64) in Stoughton and Zechner (1998). Our example here generalizes the results of those models to an arbitrary trading horizon.

Note that the agent's trading position is, in general, never equal to the competitive allocation α^p , but is constantly approaching it. If the agent trades sufficiently frequently, his trading policy converges, for all practical purposes, to the competitive allocation. We illustrate the equilibrium for a particular set of parameters below.

Example 1: Let $r = 4\%$, $\sigma = 100$, $\gamma^I r \sigma^2 = 5$, $\gamma^A r \sigma^2 = 50$, $\alpha^- = 1$, $\mu_1 = 50$, and $T = 10$ years. Suppose the agent can trade once per quarter. Then the optimal trading policy is illustrated below:



From equation (17), the agent's shareholdings α_t converge to α^P at a rate determined by the sensitivity v_{1t} of the share price to the agent's holdings. Indeed, we can write

$$\alpha_t - \alpha^P = e^{-\lambda_t \Delta} (\alpha_{t-1} - \alpha^P),$$

where the convergence rate λ_t is given by

$$\lambda_t = \Delta^{-1} \log (1 + \delta \gamma^* r \sigma^2 / v_{1t}).$$

We explore the dynamics of v_{1t} and λ_t below.

PROPOSITION 11. The share price sensitivity v_{1t} is increasing and the convergence rate λ_t is decreasing in the horizon $T - t$. As $T - t \rightarrow \infty$, $v_{1t} \rightarrow v_1^*$ and $\lambda_t \rightarrow \lambda^*$, where v_1^* is the steady state solution to (16), given by

$$v_1^* = \frac{1}{2} \left(\frac{\mu_1}{r} - \gamma^A \sigma^2 \right) + \frac{1}{2} \sqrt{ \left(\frac{\mu_1}{r} - \gamma^A \sigma^2 \right)^2 + 4 \delta \gamma^* r \sigma^2 \left(\frac{\mu_1}{r} + \gamma^I \sigma^2 \right) },$$

$$\text{and } \lambda^* = \Delta^{-1} \log (1 + \delta \gamma^* r \sigma^2 / v_1^*).$$

PROOF: Define

$$h(v_1) \equiv \delta(\mu_1 + \gamma^l r \sigma^2) + \exp(-r\Delta) \frac{v_1^2}{v_1 + \delta \gamma^* r \sigma^2}.$$

Then v_{1t} follows the difference equation $v_{1t-1} = h(v_{1t})$. Moreover, v_1^* is the unique non-negative solution to $v_1 = h(v_1)$. Note that h is continuous, $h(0) \geq 0$, and h is weakly increasing. Thus, for $0 \leq v_{1t} < v_1^*$, we have $v_{1t} < h(v_{1t}) = v_{1t-1} < v_1^*$. Thus, v_{1t} increases with $T - t$ and converges to v_1^* . The remaining results follow straightforwardly from the definition of λ_t . ♦

The next result demonstrates the effect of reducing the trading interval Δ for the agent. Smaller Δ implies a smaller “commitment” interval over which the agent will not trade. As we show below, this leads to more rapid convergence to the price-taking solution. The limiting case $\Delta = 0$ corresponds to continuous trading, where the agent has no commitment ability. As the proposition shows, the qualitative behavior of the equilibrium in this case depends critically on the sign of $z_{\alpha\alpha} = \mu_1 - \gamma^l r \sigma^2$.

PROPOSITION 12. v_1^* is increasing and λ^* is decreasing in Δ . As $\Delta \rightarrow 0$,

$$v_1^* \rightarrow \max(0, \mu_1/r - \gamma^l \sigma^2),$$

$$\lambda^* \rightarrow \gamma^* r \sigma^2 / v_1^*.$$

PROOF: v_1^* increasing follows since δ is increasing in Δ . Given that, to show λ^* is decreasing it is sufficient to show that $g = \Delta^{-1} \log(1 + \delta a)$ is decreasing in Δ for $a > 0$. Differentiating g we find that g is decreasing iff

$$(1 + \delta a) \log(1 + \delta a) > a \Delta d\delta/d\Delta.$$

Since for $x \geq 0$, $(1 + x) \log(1 + x) \geq x$, it is sufficient to show that $\delta > \Delta d\delta/d\Delta$. This can be shown to follow from the inequality, $e^x > 1 + x$.

For convergence, note that for Δ small, $\delta \approx \Delta$. Thus, v_1^* follows immediately. If $\mu_1/r - \gamma^l \sigma^2 > 0$, then

$$\Delta^{-1} \log(1 + \delta \gamma^* r \sigma^2 / v_1^*) \approx \Delta^{-1} \delta \gamma^* r \sigma^2 / v_1^* \rightarrow \gamma^* r \sigma^2 / v_1^* = \lambda^*.$$

If $\mu_1/r - \gamma^l \sigma^2 = 0$, then $v_1^* = k \delta^{1/2}$, and

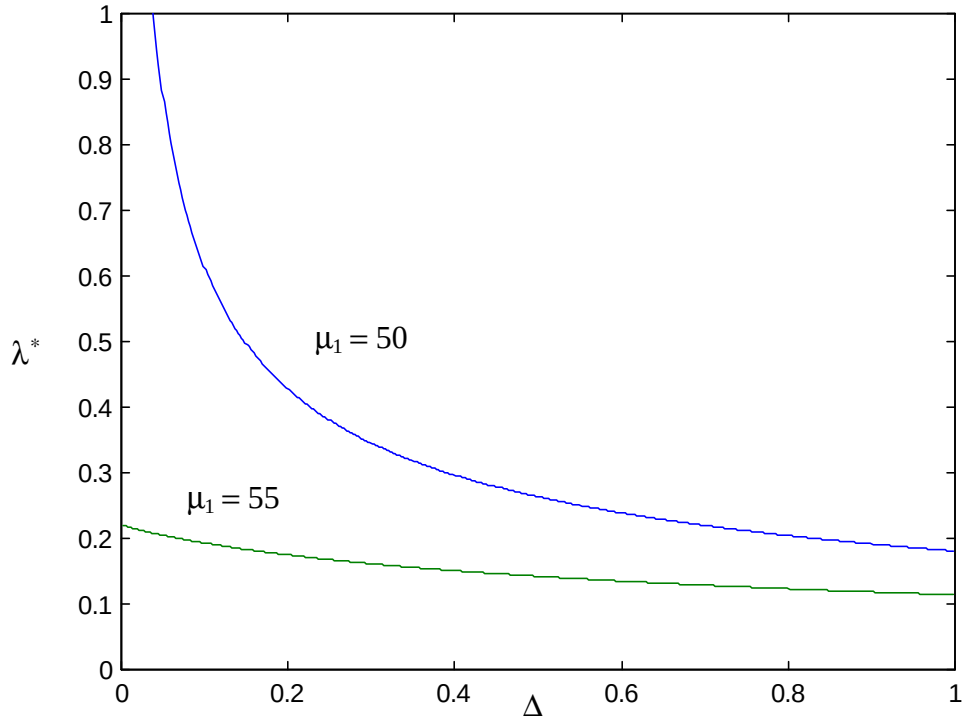
$$\Delta^{-1} \log(1 + \delta \gamma^* r \sigma^2 / v_1^*) \approx \Delta^{-1} \delta^{1/2} \gamma^* r \sigma^2 / k \rightarrow \infty = \lambda^*.$$

The case $\mu_1/r - \gamma^l \sigma^2 < 0$ follows similarly. ♦

The figure below depicts the steady-state convergence rate λ^* as a function of the time interval Δ between trading opportunities, for the parameters of Example 1. For $\mu_1 > 50$, $z_{\alpha\alpha} = \mu_1 - \gamma^l r \sigma^2 > 0$ and λ^* converges to a finite quantity. Thus, the agent trades

gradually to α^p even with continuous trading opportunities. Recall from **PROPOSITION 7** that $z_{\alpha\alpha} > 0$ implies that a Walrasian equilibrium does not exist.

For $\mu_1 \leq 50$, however, $z_{\alpha\alpha} = \mu_1 - \gamma^a r \sigma^2 \leq 0$, and α^p is a Walrasian equilibrium. In this case, $\lambda^* \rightarrow \infty$ as trading opportunities become continuous. Thus, in the limit of continuous trading, the agent trades instantly (at date 0) to α^p , and the share price sensitivity $v_1^* \rightarrow 0$. This is justified by the fact that if investors expect the agent to trade immediately to α^p , then the share price will not depend on current holdings α , and if the share price is independent of α , the agent will behave as a price-taker. Finally, note that given *any* discrete trading interval (even as short as one day) the agent will gradually (rather than immediately) trade to α^p .



5. A Continuous Time Approximation

Having found the discrete time solution for this model, we now consider a continuous time approximation to the discrete model. For simplicity, we consider the limit $T \rightarrow \infty$.⁴⁵ The main motivation for doing so is that equations (10) and (11) that describe the solution in the discrete setting are not analytically tractable except for special cases like the linear model of Section 4.1. The increased tractability of the continuous case will allow us to generalize many of the key conclusions of the previous section.

⁴⁵ The non-stationary case is equally tractable but formulas are less elegant.

Equation (10) is a difference equation describing the optimal evolution of $\alpha(t)$. In the continuous time limit, (10) becomes the differential equation

$$\alpha'(t) = \frac{z_\alpha(\alpha, t) - v(\alpha, t)}{V_\alpha(\alpha, t)}, \quad (18)$$

if $V_\alpha(\alpha, t) > 0$. If $V_\alpha \rightarrow 0$ in the continuous time limit, then $\alpha' \rightarrow \pm\infty$, and the equilibrium trading strategy involves a discrete jump from α to some α^* closer to $\alpha^p(t)$ on date t .

Thus, whether the agent trades continuously or discretely depends upon the equilibrium share price V . Recall from equation (11) derived from the equilibrium solution in **PROPOSITION 5**, that if starting from (α, t) , $\alpha(\tau)$ for $\tau \geq t$ is the optimal policy for the agent, then

$$V(\alpha(t), t) = \int_t^\infty e^{-r(\tau-t)} v(\alpha(\tau), \tau) d\tau.$$

Taking the derivative of the above with respect to t yields

$$dV/dt = r V(\alpha, t) - v(\alpha, t),$$

that is, the share price grows at the risk-free rate less the risk-adjusted dividend yield (this is also equivalent to equation (2)). Using equation (18) to substitute for $V_\alpha \alpha'$, we can rewrite this as

$$r V = dV/dt + v = V_t + V_\alpha \alpha' + v = V_t + (z_\alpha - v) + v,$$

which yields the differential equation

$$V_t(\alpha, t) = r V(\alpha, t) - z_\alpha(\alpha, t). \quad (19)$$

Note that (19) holds whenever the agent trades continuously; i.e., whenever $\alpha(t) = \alpha(t-)$. For a given α , if this is true for all t , then the equilibrium share price is given by

$$V(\alpha, t) = \int_t^\infty e^{-r(s-t)} z_\alpha(\alpha, s) ds. \quad (20)$$

Thus, equation (20) provides an explicit expression for $V(\alpha, t)$ in terms of the primitives of the model. Note in particular that it does not depend on the agent's trading strategy; α is held constant in the integration. Given this expression for V , equation (18) yields a differential equation for the agent's trading strategy $\alpha(t)$. We now verify that this indeed represents an equilibrium with continuous trading.

PROPOSITION 13. Suppose V is given by (20) and α satisfies (18). Then if $V_\alpha > 0$, α is an optimal no-commitment trading strategy for the agent and V is the equilibrium share price. The agent's certainty equivalent payoff given shareholdings α as of date t is given by

$$Z(\alpha, t) = \int_t^\infty e^{-r(s-t)} z(\alpha, s) ds. \quad (21)$$

PROOF: For any continuous (and bounded variation) strategy α , recall the agent's certainty equivalent payoff at date t is given by

$$\int_t^\infty e^{-r(\tau-t)} [z(\alpha(\tau), \tau) d\tau - V(\alpha(\tau), \tau) d\alpha(\tau)].$$

From (20), this becomes

$$\int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau), \tau) d\tau - \int_t^\infty e^{-r(\tau-t)} \left[\int_\tau^\infty e^{-r(s-\tau)} z_\alpha(\alpha(\tau), s) ds \right] d\alpha(\tau).$$

Exchanging the order of integration,

$$\int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau), \tau) d\tau - \int_t^\infty e^{-r(s-t)} \left[\int_t^s z_\alpha(\alpha(\tau), s) d\alpha(\tau) \right] ds,$$

or,

$$\begin{aligned} & \int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau), \tau) d\tau - \int_t^\infty e^{-r(s-t)} [z(\alpha(s), s) - z(\alpha(t), s)] ds \\ &= \int_t^\infty e^{-r(s-t)} z(\alpha(t), s) ds. \end{aligned}$$

Thus, the agent's certainty equivalent payoff is path-independent and hence maximized by the strategy implied by (18). To see that jumps are not optimal, suppose that $\alpha(t) \neq \alpha(t-)$. In this case the certainty equivalent is given by

$$(\alpha(t-) - \alpha(t)) V(\alpha(t), t) + Z(\alpha(t), t).$$

This has derivative with respect to $\alpha(t)$ of

$$(\alpha(t-) - \alpha(t)) V_\alpha(\alpha(t), t),$$

where we use the fact that $Z_\alpha = V$. Given $V_\alpha > 0$, $\alpha(t) = \alpha(t-)$ is optimal for all t , confirming (21).

Next we verify that V is the correct equilibrium price. Recall that

$$V(\alpha(t), t) = \int_t^\infty e^{-r(s-t)} v(\alpha(s), s) ds.$$

From (18), we can rewrite this as

$$\int_t^\infty e^{-r(s-t)} [z_\alpha(\alpha(s), s) - V_\alpha(\alpha(s), s) \alpha'(s)] ds = \int_t^\infty e^{-r(s-t)} z_\alpha(\alpha(t), s) ds,$$

where we use the same simplification as in the derivation of Z , with z_α and V_α in place of z and V . Alternatively, it follows immediately from **PROPOSITION 9**, since $Z_\alpha(\alpha, t) = V(\alpha, t)$ given the optimal policy is continuous. ♦

Note that equation (20) is quite remarkable. It states that the equilibrium share price can be “decoupled” from the solution to the optimal trading strategy and solved for directly from the model primitives. It also has a strikingly simple interpretation. Recall from **PROPOSITION 4** that

$$z_\alpha(\alpha, t) = \mu(\alpha, t) - \alpha \gamma^\rho r \sigma^2(\alpha, t) + b(\alpha, t).$$

Thus, the current share price is equal to the present value of the expected dividends, **under the assumption that the agent does not trade**, and adjusted for the risk premium and control benefits of the agent. In the time-stationary case without control benefits, this reduces to the annuity value of the current dividend rate, adjusted by the agent's risk premium. If the agent's position α is large, share prices will look as though they reflect a "too large" risk premium relative to investors' risk aversion.

Of course, the assumption that the agent does not trade is incorrect. In equilibrium, the agent holding a large position will sell shares, and effort and hence dividends will fall. The result shows, however, that rather than solving the complicated problem of anticipating the rate of sale of the agent and the corresponding change in the dividends, it is sufficient to simply adjust the present value of the "naïve" forecast of dividends using the agent's risk premium rather than the markets.

PROPOSITION 13 requires $V_\alpha > 0$ for V defined by (20). A sufficient condition for this is $z_{\alpha\alpha} > 0$, precisely the condition that implies the non-existence of the Walrasian equilibrium, and the condition that was critical for determining the limiting behavior of the linear example of Section 4.1 as shown in **PROPOSITION 12**. This suggests that when this condition is violated, the equilibrium in continuous time will involve discontinuous jumps in the holdings, as was true in the example of Section 4.1. We establish this below:

PROPOSITION 14. Suppose $z_{\alpha\alpha} \leq 0$. Then the optimal trading strategy in the continuous-time limit coincides with the Walrasian equilibrium: the agent trades immediately to $\alpha^p(t)$.

PROOF: Suppose $V_\alpha(\alpha, t) > 0$ so that the agent trades continuously from α at date t . Then from (19), $V_{\alpha t} = r V_\alpha - z_{\alpha\alpha} > 0$, so that the agent trades continuously from α for all $s > t$. But then (20) implies that $V_\alpha(\alpha, t) \leq 0$. Thus, $V_\alpha(\alpha, t) = 0$ for all (α, t) , and so the agent behaves as a price-taker. Thus, the equilibrium corresponds to the Walrasian equilibrium, and the agent jumps immediately to $\alpha^p(t)$. ♦

The last two results thus generalize the intuition of the earlier examples: when the Walrasian equilibrium exists, it coincides with the time-consistent optimal trading strategy when the time between trading opportunities goes to zero. When the Walrasian equilibrium fails to exist, the optimal trading strategy approaches the price-taking level gradually over time.^{46,47}

⁴⁶ This is also consistent with the conjecture of Admati, et al. (1994), that when the agent can trade infinitely often, the equilibrium will coincide with the Walrasian equilibrium (which they refer to as the "globally stable allocation"). Their model is inherently static, however, and they do not have a prediction regarding equilibrium when the Walrasian equilibrium does not exist.

⁴⁷ The equilibria identified here in Proposition 13 and **PROPOSITION 14** correspond to limits of the unique subgame perfect equilibrium of the discrete trading case. As such they are the natural equilibria to focus on. However, in the continuous trading setting, other equilibria are likely to emerge. See, for example, DeMarzo and Bizer (1993) and Ausubel and Deneckere (1989) for examples of the type of

5.1. Example: Linear Returns

In this section we consider again the linear example of Section 4.1, now working directly from the continuous time solution derived above. In that example,

$$z_\alpha(\alpha, t) = \mu(\alpha, t) - \alpha \gamma^a r \sigma^2 = \mu_0(t) + \mu_1 \alpha - \alpha \gamma^a r \sigma^2.$$

Thus, from (20),

$$V(\alpha, t) = \int_t^\infty e^{-r(\tau-t)} \mu_0(\tau) d\tau + \frac{1}{r} (\mu_1 - \gamma^a r \sigma^2) \alpha.$$

That is,

$$V(\alpha, t) = v_1(t) \alpha + v_2(t),$$

with

$$v_1(t) = \frac{1}{r} (\mu_1 - \gamma^a r \sigma^2) \equiv v_1 \text{ and } v_2(t) = \int_t^\infty e^{-r(\tau-t)} \mu_0(\tau) d\tau.$$

Recall that we require $V_a > 0$ or $v_1(t) > 0$. Thus, this solution is valid only if

$$\mu_1/r - \gamma^a \sigma^2 > 0.$$

Note that this is precisely the condition $z_{\alpha\alpha} > 0$. Just as in **PROPOSITION 12**, this condition determines when the optimal trading strategy is continuous in the limit. When it does not hold, the agent trades immediately to α^p .

Next, from (18) we have

$$\alpha' = (z_\alpha - v) / v_1 = -(\alpha - \alpha^p) \gamma^* r \sigma^2 / v_1. \quad (22)$$

Thus, the rate of convergence is given by

$$\lambda^* = \gamma^* r \sigma^2 / v_1 = \gamma^* r \sigma^2 / (\mu_1/r - \gamma^a \sigma^2). \quad (23)$$

This coincides precisely with the stationary discrete time case ($T = \infty$) of **PROPOSITION 12**. From (22), the optimal trading policy is easily determined to be:

$$\alpha(t) = \alpha^p + (\alpha^- - \alpha^p) \exp(-\lambda^* t). \quad (24)$$

5.2. General Stationary Case

The previous results have established that the character of the equilibrium depends critically on the convexity or concavity of z . When z is neither convex nor concave, the equilibrium will involve both jumps as well as gradual trading. Solving for the general case is complicated. Rather than attempt that here, we now demonstrate the key ideas for the simpler stationary case.

Suppose $z(\alpha, t)$ and $\alpha^p(t)$ are independent of t , and so we write them simply as $z(\alpha)$ and α^p . If the agent trades continuously at α , then (20) implies

multiplicity that is possible in these models. We view the continuous time setting as an approximation of the discrete case (in reality trading is discrete), and thus we do not explore these other equilibria.

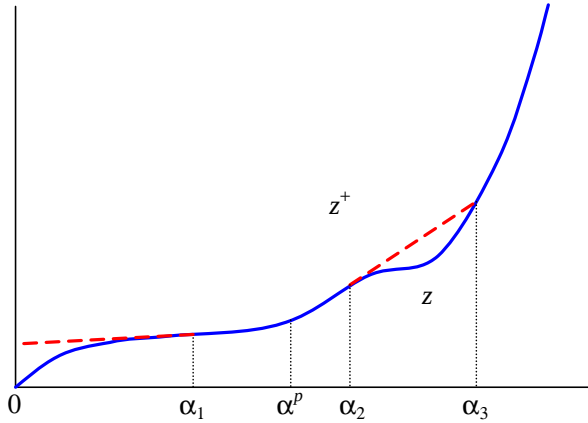
$$V(\alpha) = \frac{z_\alpha(\alpha)}{r}.$$

Since only points with $V_\alpha \geq 0$ can occur along the equilibrium path, at points such that $z_{\alpha\alpha}(\alpha) \leq 0$, the agent will jump towards α^p . This suggests that the equilibrium can be described by an appropriate convexification z^+ of z , which we define as follows:

1. $z^+(\alpha) \geq z(\alpha)$,
2. $z^+(\alpha)$ is convex in α ,
3. $z^+(\alpha^p) = z(\alpha^p)$,
4. if $z^+(\alpha) > z(\alpha)$, then there exists $\alpha^* \in (\alpha, \alpha^p]$ such that for $\hat{\alpha} \in [\alpha, \alpha^*]$,

$$z^+(\hat{\alpha}) = z(\alpha^*) + (\hat{\alpha} - \alpha^*) z_\alpha(\alpha^*). \quad (25)$$

See the example in the figure below:



$$\alpha^*(\alpha) = \begin{cases} \alpha_1 & \text{for } \alpha \leq \alpha_1 \\ \alpha_2 & \text{for } \alpha_2 \leq \alpha < \alpha_3 \\ \alpha & \text{otherwise} \end{cases}$$

In terms of the agent's optimal strategy, the agent trades continuously ($\alpha(t) = \alpha(t-)$) at points for which $z^+(\alpha) = z(\alpha)$, but otherwise will trade a large block and jump to the holdings $\alpha^*(\alpha)$, defined by 4 above.

Given this, we extend the definition of the share price (20) in this case as follows:

$$V(\alpha) = \frac{z_\alpha^+(\alpha)}{r}. \quad (26)$$

Note that (26) coincides with (20) along the equilibrium path, and since z^+ is convex, $V_\alpha \geq 0$.⁴⁸

⁴⁸ Note that z^+ need not be everywhere differentiable (consider for example the point α_3 in the example above). Letting $z_\alpha^+(\alpha^+)$ and $z_\alpha^+(\alpha^-)$ be the right and left subgradients of z_α^+ (which always exist since z^+ is convex), we then interpret (26) as $V(\alpha) \in [z_\alpha^+(\alpha^-), z_\alpha^+(\alpha^+)] / r = [V(\alpha^-), V(\alpha^+)]$, which defines a range of possible values for V . In this case, the agent's holdings remain fixed at α from t to $t + \Delta$ as the price gradually moves through the range, where Δ is defined such that

$$V(\alpha^+) = \int_t^{t+\Delta} e^{-r(s-t)} V(\alpha, s) ds + e^{-r\Delta} V(\alpha^-),$$

if $\alpha > \alpha^p$. If $\alpha < \alpha^p$, the roles of α^+ and α^- are reversed.

Recall from **PROPOSITION 13** that when the agent's trading strategy is continuous, the agent's certainty equivalent payoff is given by $Z(\alpha, t)$, which in this case is $z(\alpha) / r$. If the agent jumps to α^* , the corresponding certainty equivalent is

$$\frac{z(\alpha^*(\alpha))}{r} + (\alpha - \alpha^*(\alpha)) V(\alpha^*(\alpha)) = \frac{z(\alpha^*(\alpha)) + (\alpha - \alpha^*(\alpha)) z_\alpha(\alpha^*(\alpha))}{r} = \frac{z^+(\alpha)}{r},$$

since $V(\alpha^*) = z_\alpha(\alpha^*) / r$. This suggests that the correct extension of the agent's certainty equivalent in this case is given by

$$Z(\alpha) = \frac{z^+(\alpha)}{r}.$$

The following result confirms this characterization of the equilibrium in the limit of continuous trading:

PROPOSITION 15. Suppose V is given by (26) and α satisfies (18), with discontinuities given by α^* . Then α is an optimal no-commitment trading strategy for the agent and V is the equilibrium share price. The agent's certainty equivalent payoff given shareholdings α is given by $Z(\alpha)$.

PROOF: For any strategy α (with bounded variation), recall the agent's certainty equivalent payoff at date t is given by

$$\int_t^\infty e^{-r(\tau-t)} [z(\alpha(\tau)) d\tau - V(\alpha(\tau)) d\alpha(\tau)].$$

From (26), this becomes

$$\int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau)) d\tau - \int_t^\infty e^{-r(\tau-t)} \left[z_\alpha^+(\alpha(\tau)) \int_\tau^\infty e^{-r(s-\tau)} ds \right] d\alpha(\tau).$$

Exchanging the order of integration,

$$\int_t^\infty e^{-r(\tau-t)} z(\alpha(\tau)) d\tau - \int_t^\infty e^{-r(s-t)} \left[\int_t^s z_\alpha^+(\alpha(\tau)) d\alpha(\tau) \right] ds. \quad (27)$$

Now, since z^+ is convex,

$$\int_t^s z_\alpha^+(\alpha(\tau)) d\alpha(\tau) \geq z^+(\alpha(s)) - z^+(\alpha(t)),$$

with equality if and only if α is discontinuous only over regions for which z^+ is linear. Thus, (27) can be rewritten,

$$\frac{z^+(\alpha(t))}{r} + \int_t^\infty e^{-r(\tau-t)} [z(\alpha(\tau)) - z^+(\alpha(\tau))] d\tau - B,$$

where $B \geq 0$, and $B = 0$ if and only if α is discontinuous only over regions for which z^+ is linear. Finally, since $z^+ \geq z$, the strategy α is optimal if it is discontinuous when $z^+ \neq z$, and only over regions where z^+ is linear. These are precisely the properties of α^* . In this case, the certainty equivalent is given by Z .

From this, the share price V follows from **PROPOSITION 9** since

$$V(\alpha) = V(\alpha^*(\alpha)) = Z_\alpha(\alpha) = z^+_\alpha(\alpha) / r. \quad \blacklozenge$$

6. IPOs and Lockups

Thus far, we have not considered the origins of the large shareholder in our model. Indeed, since the large shareholder always trades towards α^p , the dynamics of our model does not lead naturally to the creation of large positions.

One possible source for the emergence of large shareholders is through time variation in $\alpha^p(t)$. For example, suppose that for short periods of time, there are large control benefits associated with share ownership (this might occur, for example, during a control contest). Then (8) implies that during that period, $\alpha^p(t)$ is large. Thus, even starting from low initial ownership, the large shareholder will accumulate shares during such a period. Once the control benefits dissipate, $\alpha^p(t)$ will fall back to the optimal risk sharing solution, and the large shareholder will gradually unwind the position. Thus, time-varying control benefits lead to non-monotone dynamics of the large shareholder's position.

Another natural source for the creation of the large shareholder is the firm itself. Since the share price is increasing in the large shareholder's stake, the original owners of the firm have an incentive to maximize this stake. While the large shareholder would not voluntarily accumulate shares above α^p if they must be acquired in the market, the firm can subsidize the large shareholder by offering shares at a below market price. Given the externalities of the large shareholder's monitoring activity, such subsidization can make both the initial owners and the remaining shareholders of the firm better off.

One obvious example of subsidization of large shareholders is the use of stock grants for management. By granting stock at a below market price, management's incentives for effort provision are enhanced. A second important example occurs during IPO's. There, it is not unusual for large blocks of shares to be awarded to institutional holders. If these institutional holders perform a monitoring function for the firm, the creation of these blocks enhances the value of the firm.

As a simple example, consider the case in which at the IPO, the original owners of the firm are selling shares to both a large institutional investor as well as small individual investors. Let the IPO price be denoted p . Then the payoff to the large shareholder from acquiring α shares is

$$Z(\alpha, 0) - \alpha p,$$

where Z is the certainty equivalent (9). Note first that since Z is convex (see **PROPOSITION 9**), given a fixed price the large shareholder will generally not choose an interior allocation.⁴⁹

⁴⁹ The exception to this is the special case in which Z is linear. This only occurs if trading is continuous and the Walrasian equilibrium exists. In that case, the large shareholder immediately trades to α^p after acquiring the initial holdings, and so there is no benefit to subsidizing the large shareholder's initial stake.

Suppose instead that the underwriter can set an IPO price, and then choose the allocation for the large shareholder and the small investors subject to their participation constraints. We do not allow the underwriter to price discriminate between the small and large shareholders. This is essentially the mechanism proposed in Stoughton and Zechner (1998). In this case the underwriter's problem can be written as

$$\begin{aligned} \max_{\alpha, p} \quad & p \\ \text{s.t.} \quad & \text{(LP)} \quad Z(\alpha, 0) - \alpha p \geq Z(0, 0), \\ & \text{(SP)} \quad p \leq V(\alpha_0(\alpha), 0) \\ & \text{(LC)} \quad \alpha \leq \bar{\alpha}. \end{aligned} \tag{28}$$

The constraints (LP) and (SP) represent the large shareholder's and the small investor's participation constraints, respectively. Note that the small shareholders constraint is based on the market price of the shares *once the market opens* and the large shareholder trades to $\alpha_0(\alpha)$. Lastly, we have included a "liquidity constraint" (LC), that represents (un-modeled) restrictions on the size of the large shareholder's position.⁵⁰

We have the following result:

PROPOSITION 16. At the solution to the IPO problem (28), constraint (LP) always binds and $\alpha = \bar{\alpha}$ is optimal. Unless trading is continuous and $\alpha_0(\bar{\alpha}) = \alpha_0(0) = \alpha^P(0)$, there is IPO underpricing: $p < V(\alpha_0(\alpha), 0)$.

PROOF: The large shareholder's participation constraint can be rewritten as

$$p \leq \frac{Z(\alpha, 0) - Z(0, 0)}{\alpha}. \tag{29}$$

Also, from **PROPOSITION 9**, $V(\alpha_0, 0) = Z_{\alpha}(\alpha, 0)$. Thus, (LP) implies (SP) since the convexity of Z implies

$$Z_{\alpha}(\alpha, 0) \geq \frac{Z(\alpha, 0) - Z(0, 0)}{\alpha},$$

and the inequality is strict (and there is underpricing) as long as Z is not linear in the range $[0, \alpha]$. Z linear can occur only if trading is continuous, and it implies that $\alpha_0(\bar{\alpha}) = \alpha_0(0) = \alpha^P(0)$. Finally, since (29) is increasing in α for Z not linear, $\bar{\alpha}$ is optimal. ♦

As a simple example, consider the linear model of Section 5.1. In this case, given $\mu_1/r - \gamma^p \sigma^2 > 0$,

$$Z(\alpha, 0) = \alpha \int_0^{\infty} e^{-r(\tau-t)} \mu_0(\tau) d\tau + \frac{1}{2} \alpha^2 (\mu_1/r - \gamma^p \sigma^2).$$

Thus, the IPO price is

⁵⁰ These restrictions may correspond to portfolio constraints on the part of the large shareholder, market liquidity concerns, corporate control concerns of management, etc.

$$p = \int_0^\infty e^{-r(\tau-t)} \mu_0(\tau) d\tau + \frac{1}{2} \bar{\alpha} (\mu_1 / r - \gamma^a \sigma^2).$$

Since $V = Z_\alpha$, the amount of IPO underpricing is

$$V(\bar{\alpha}, 0) - p = \frac{1}{2} \bar{\alpha} (\mu_1 / r - \gamma^a \sigma^2).$$

In this case, 50% of the gains associated with the presence of the large shareholder are dissipated through under-pricing. The remainder are captured by the initial owners of the firm.⁵¹ Note also that the underpricing is increasing in μ_1 and decreasing in γ^a . These two unobserved parameters have the opposite impact on the speed of adjustment λ from (23). Thus, this model has the empirical prediction that the degree of underpricing will be negatively correlated with the speed of adjustment of the large shareholder's position.⁵²

It is also interesting to consider the effect of trading restrictions in this setting. Consider the possibility of a lockup arrangement that allows the agent to commit to an equity stake for some period of time. How would a lockup provision affect company value and the agent's stake after the IPO?

Given continuous trading on $[0, \infty)$, consider a lockup provision that prohibits the agent from trading on $(0, t)$. That is, starting with pre-IPO holdings of α^- , the agent can trade at the IPO (date 0), but then cannot trade again until date t . Intuitively, if the environment is stationary so that there is no "efficiency" motive for dynamic trading, this opportunity to commit not to trade should be valuable for the agent. We verify this below for the linear example of Section 5.1, and show also that it leads to a higher share price and higher long run holdings by the large shareholder.

PROPOSITION 17. Consider the linear model of Section 5.1, with z convex and $\alpha^- > \alpha^p$. Then utility of the large shareholder and the share price are increasing in the lockup period t , while the initial holdings $\alpha^L(0)$ are decreasing in t . On the other hand, for $\tau > t$, $\alpha^L(\tau)$ is increasing in t , so that the lockup increases the large shareholder's long run holdings.

PROOF: With the lockup, the agent's optimal trading decision at the IPO stage is given by

$$\max_{\alpha_0} V^L(\alpha_0)(\alpha^- - \alpha_0) + \delta_0 z(\alpha_0) + e^{-rt} Z(\alpha_0).$$

Given our assumptions $Z(\alpha_0) = z(\alpha_0)/r$, and so this is equivalent to,

$$\frac{1}{r} \max_{\alpha_0} r V^L(\alpha_0)(\alpha^- - \alpha_0) + z(\alpha_0). \quad (30)$$

Using (20) and **PROPOSITION 5**,

⁵¹ The large shareholder does not capture any of the gains since the constraint (LP) implicitly assumes that the underwriter has all of the bargaining power. If large shareholders are "scarce" and this is not the case, this will lower the price further and increase the amount of underpricing.

⁵² This relationship is not limited to this linear example. In general, the degree of underpricing is positively related, and the speed of adjustment is negatively related, to the convexity of Z .

$$r V^L(\alpha_0) = l v(\alpha_0) + (1 - l) z_\alpha(\alpha_0),$$

where $l = 1 - e^{-rt} \approx r t$. Since $(v(\alpha_0) - z_\alpha(\alpha_0))(\alpha^- - \alpha_0) > 0$ from **PROPOSITION 9**, (30) is increasing in l , so that the large shareholder's payoff is increasing in t .

Solving (30), we get

$$\alpha_0^L = \frac{\alpha^-(\mu_1 - \gamma^a r \sigma^2 + l \gamma^* r \sigma^2) + l \gamma^l r \sigma^2}{\mu_1 - \gamma^a r \sigma^2 + 2l \gamma^* r \sigma^2}, \quad (31)$$

so that

$$\frac{\partial}{\partial l} \alpha_0^L = -\frac{(\alpha^- - \alpha^p)(\mu_1 - \gamma^a r \sigma^2) \gamma^* r \sigma^2}{(\mu_1 - \gamma^a r \sigma^2 + 2l \gamma^* r \sigma^2)^2} < 0,$$

and

$$\frac{\partial}{\partial l} r V^L(\alpha_0^L) = \frac{2l(\alpha^- - \alpha^p)(\mu_1 - \gamma^a r \sigma^2 + l \gamma^* r \sigma^2)(\gamma^* r \sigma^2)^2}{(\mu_1 - \gamma^a r \sigma^2 + 2l \gamma^* r \sigma^2)^2} > 0.$$

Finally, consider lockup periods $t > t'$ and corresponding trading strategies α^L and $\alpha^{L'}$. Since the initial share price is increasing, but the initial holdings are decreasing in t , it must be the case that $\alpha^L(t) > \alpha^{L'}(t)$, since otherwise $\alpha^L \leq \alpha^{L'}$ by the monotonicity result of **PROPOSITION 9**. Also by monotonicity, the same result holds for $\tau > t$. ♦

Which types of firm benefit the most from lockup restrictions? Given (31) above, one can calculate the share price benefit of the lockup as,

$$V^L(\alpha_0^L) - V(\alpha^-) = \frac{(l \gamma^* r \sigma^2)^2 (\alpha^- - \alpha^p)}{r [\mu_1 - \gamma^a r \sigma^2 + 2l \gamma^* r \sigma^2]}.$$

This benefit is decreasing in μ_1 and increasing in γ^a , as is the speed of adjustment λ (from (23)).⁵³ (A similar result holds for the welfare of the large shareholder.) Thus, holding fixed the (un-modeled) cost of the lockup restriction, we have the empirical prediction that the presence of lockup restrictions should be positively correlated with the adjustment speed of the large shareholder post lockup.

7. Further Extensions of the Model

Agent Control of Volatility. Our examples thus far were such that the agent influences the mean but not the volatility of the dividend process. As we mentioned in the introduction, one of the advantages of our approach is that it allows us to analyze models where volatility can be impacted as well. For example, suppose that the agent engages in risk management activities and so only affects the volatility of the cash flows. That is, let

⁵³ The second claim requires some calculation, since both γ^* and α^p depend on γ^a .

$\mu(\alpha) = \mu_0$ and $\sigma^2(\alpha) = \bar{\sigma}^2 / \alpha^{1+c}$ for some $c > 0$. If $\gamma^I > 0$, the sufficient conditions for existence and the uniqueness of the equilibrium are met, and, in the continuous trading limit, the share price and trading policy are given by:

$$V(\alpha) = \frac{\mu_0}{r} - \alpha \gamma^a \bar{\sigma}^2(\alpha), \quad \frac{d\alpha}{dt} = \frac{r\gamma^*}{c\gamma^a}(\alpha^p - \alpha).$$

That is, the agent's stake decays exponentially towards α^p with adjustment speed $\lambda = \frac{r\gamma^*}{c\gamma^a}$. Intuitively, higher c implies a higher sensitivity of volatility – and hence share price – to α , which slows the speed of adjustment. As a second example, consider a model where both mean and volatility are impacted, with $\mu(\alpha) = \mu_0 + \mu_1 \alpha$, $\mu_1 > 0$, $\sigma^2(\alpha) = \bar{\sigma}^2 / \alpha$, and $\gamma^I > 0$. In this case, the optimal trading policy is not analytically tractable. But as $\gamma^I \rightarrow 0$, the optimal trading policy approaches a sell-off strategy that is linear in time:

$$\alpha(t) = \alpha_0 - \frac{\gamma^* r \bar{\sigma}^2}{\mu_1 / r} t.$$

Alternative Motives for Trade. So far we have interpreted the model as though the motive for trade is improved diversification. We could, alternatively, interpret it as a tradeoff resulting liquidity needs or tax differences.⁵⁴ For example, consider the situation in which investors value the dividend as $\mu(\alpha)$, but the agent values it as $(1-\kappa)\mu(\alpha)$. Here, $\kappa \in [0,1]$ is the agent's tax disadvantage (or “cost” associated with liquidity needs). Let the agent and investors be risk neutral, or let $\sigma = 0$, so that there is no diversification motive. For simplicity, let $b=0$, and $\mu(\alpha)=\mu_1 \alpha$, with $\mu_1>0$. Using our results, we have immediately,

$$\alpha^s = \frac{1}{1+\kappa}, \quad \alpha^c = \frac{\alpha^-}{1+\kappa} \leq \alpha^s, \quad \alpha^p = 0$$

$$V(\alpha, t) = (1-\kappa) \frac{\mu_1}{r} \alpha$$

The time-consistent solution is given by (16) with the speed of convergence replaced by $\lambda^* = r \kappa / (1-\kappa)$. An increase in κ raises the speed of convergence to the competitive solution. This explains why the market value of the asset is decreasing in κ (recall that investors do not pay κ , yet it appears in the expression for V). Thus, risk aversion is not necessary in order to provide the agent with an incentive to trade: all that is really needed is a wedge between the agent's and the investors' marginal benefit of holding the stock.⁵⁵

⁵⁴ We thank D. Duffie for this interpretation.

⁵⁵ If the agent and the investors are, in addition, risk averse, the optimal trading policy has to take into account both the liquidity and the diversification motive to trade. The competitive allocation, in our example, becomes $\alpha^p = \gamma^I r \sigma^2 / (\kappa \mu_1 + \gamma^* r \sigma^2)$; i.e., the introduction of the liquidity “cost” lowers the agent's competitive allocation.

Costless Effort. Thus far we have interpreted the model in terms of an agency relationship between the large shareholder and the firm. As such we defined the model originally in terms of $(\hat{\mu}, \hat{\sigma}, f)$, where $(\hat{\mu}, \hat{\sigma})$ specify the dividend process as a function of effort, and f represents the cost of effort to the large shareholder. In Section 3.3 we then showed how this was equivalent to a reduced form specified by (μ, σ, b) , which capture the indirect effect of shareholdings on dividends and the marginal private benefit of holding shares to the large shareholder. From this reduced form, the large shareholder's flow of payoffs z is defined by

$$z_{\alpha}(\alpha, t) = \mu(\alpha, t) - \alpha \gamma^a r \sigma^2(\alpha, t) + b(\alpha, t).$$

Here we note that we can extend our model outside the moral hazard framework quite easily. Consider instead taking as primitive (μ, σ, z) . As before, (μ, σ) indicate the effect of the large shareholder's position on the dividend process of the firm. Now, z represents the *exogenous* flow of benefits to the large shareholder given his holdings. Given (μ, σ, z) , we can then define

$$b(\alpha, t) = z_{\alpha}(\alpha, t) - \mu(\alpha, t) + \alpha \gamma^a r \sigma^2(\alpha, t).$$

Given this definition of b , the results of the model can then be applied to the new setting.

For example, consider a large shareholder who can improve the dividend stream *without* having to pay an effort cost. He can either suggest dividend improving measures to management, or block dividend reducing actions favored by management (such as the adoption of entrenchment-enhancing poison pills). Neither takes costly effort on his part, but his influence on the board and management is an increasing function of his stake α .⁵⁶

In this case the large shareholder's certainty equivalent profit flow z is given by

$$z(\alpha, t) = \alpha \mu(\alpha, t) - \frac{1}{2} \alpha^2 \gamma^a r \sigma^2(\alpha, t),$$

so that

$$z_{\alpha}(\alpha, t) = \mu - \alpha \gamma^a r \sigma^2 + \alpha \mu_{\alpha} - \frac{1}{2} \alpha^2 \gamma^a r \sigma^2_{\alpha}.$$

Therefore, this is equivalent to our model with a private benefit function

$$b(\alpha, t) = \alpha \mu_{\alpha} - \frac{1}{2} \alpha^2 \gamma^a r \sigma^2_{\alpha}.$$

In essence, the private benefit is set to offset the effort cost that is assumed in the base case of our model. As an example, in the linear model setting of Section 4.1, this yields a linear private benefit⁵⁷

$$b(\alpha, t) = \mu_1 \alpha.$$

⁵⁶ Of course, if the large shareholder's suggestions are value enhancing, the fact that management did not implement them implies that there is another agency problem present between management and shareholders that the large shareholder is helping to overcome. The important point is that in this case the large shareholder does not bear a cost of making the changes, rather management suffers (presumably through reduced perks).

⁵⁷ Recall that in this example, $\hat{\mu}(e, t) = \mu_0(t) + e$ and $f(e, \alpha, t) = e^2 / (2 \mu_1)$. Thus, optimal $e = \mu_1 \alpha$ with effort cost $f = \frac{1}{2} \mu_1 \alpha^2$. The marginal effort cost is therefore $\mu_1 \alpha$, and so b offsets this.

In this case, α^p is given by the largest α such that $z_\alpha \geq v$ and so

$$\alpha^p = \gamma^l / \max[\gamma^l, \gamma^* - \mu_1 / (r \sigma^2)],$$

which is higher than in the case of costly effort by the large shareholder (there, $\alpha^p = \gamma^l / \gamma^*$). Note also that in this case $z_{\alpha\alpha} = 2 \mu_1 - \gamma^l r \sigma^2$. Using the results of Section 5.1, gradual (rather than immediate) convergence to α^p occurs if $\mu_1 > \frac{1}{2} \gamma^l r \sigma^2$, with a convergence rate of $\lambda^* = (\gamma^* r \sigma^2 - \mu_1) / (2\mu_1/r - \gamma^l \sigma^2)$. This is a slower convergence rate than when the agent must exert costly effort.

Other Settings of “Durable Goods Monopoly.” More broadly, our results can be generalized to encompass other settings outside the large shareholder framework. The system of equations (10) and (11) in the discrete-time case and (18) and (20) for the continuous-time setting depend only on (z, v) . By an appropriate choice of z , the payoff flow to the monopolist agent, and v , the common payoff flow to competing buyers, we can derive the dynamics for a large class of “durable goods monopoly” problems in settings for which buyers have a common value.

For example, consider a monopolist selling a fixed supply of a durable good for which there is a competitive rental market. Then $(1-\alpha)$ represents the quantity sold by the monopolist, and $v(\alpha, t) = R(1-\alpha)$ represents the competitive rental rate given supply $(1-\alpha)$ on date t , where $R(q)$ is the demand curve in the rental market. Finally, assume that the all production costs of the monopolist are financed using perpetual debt, so that $z(\alpha, t) = -r C(1-\alpha)$, where $C(q)$ is the total cost of producing the quantity q (so that $C'(q)$ is the marginal cost). Then our results imply that the monopolist will ultimately sell the competitive quantity, defined by $C'(q) = R(q)/r$. With continuous trading, these sales will occur gradually if $z_{\alpha\alpha} > 0$, or equivalently, if the marginal cost of production is decreasing. In this case the monopolist earns zero profits. If the marginal cost is increasing, the competitive quantity will be sold immediately.⁵⁸

8. Conclusions and Future Research

In this paper, we analyze the optimal trading and consumption policy of a large shareholder/manager of a publicly traded company whose holdings can affect company’s future dividends and, therefore, the stock price. Our model is a dynamic version of Admati et al (1994) restricted to one risky asset, and is close in spirit to the dynamic model of Kihlstrom (1998).

In an economy where the large shareholder/manager can take effort to affect the mean and variance of normally distributed future dividends, and both the large shareholder and the representative investor have CARA preferences, we formulate, and solve, a multi-period (dynamic) model for the optimal trading, consumption policy, and the equilibrium

⁵⁸ This generalizes prior literature; for example, Stokey (1981) shows that with constant marginal costs the monopolist trades immediately in the continuous time limit. The “gap case” considered by Gul, et al. (1985) is an example of increasing marginal cost.

share price. We consider the case in which the large shareholder can commit to a trading strategy, and the case in which such commitment is impossible. When commitment is possible, our results generalize several previously known static results. Noting that the commitment strategy is time-inconsistent, we show that the time-consistent strategy is to sell off the initial stake in the company until the competitive level is reached. In the continuous time limit, we obtain a closed form solution for the equilibrium share price in terms of the primitives of the model, and a differential equation for the optimal share holdings that can be explicitly solved in some cases of interest. Finally, we demonstrate a number of applications of the results.

Our paper contributes to the literature in several ways:

- a) *It is one of the first dynamic securities pricing models with moral hazard.* While CARA/normal set-up is chosen for tractability, our model allows for the manager's effort and shareholding choices to affect both the drift and volatility of the dividend process in a very general way. Share prices are not set exogenously, but determined as rational expectations, competitive equilibrium in which shareholders anticipate the manager's future trading and effort choices.
- b) *It explicitly demonstrates a variant of Coase's Conjecture in the securities market setting.* Coase (1972) conjectures that, in a dynamic setting, the price that a durable goods monopolist can charge drops to the competitive level. In the context of securities markets, where company shares play the role of "durable goods," our paper demonstrates the way in which such adjustment occurs: in the continuous trading approximation, the asymptotic adjustment towards the competitive allocation is gradual when the large shareholder's marginal benefits are increasing in the large shareholder's stake, and immediate when they are decreasing.
- c) *It leads to tractable and intuitively appealing results, especially in the continuous time approximation.* In the continuous time approximation, the problem of determining the equilibrium share price decouples from the problem of determining the optimal trading policy. As a result, our model provides a surprising simple, closed-form expression for the equilibrium share price in terms of the *primitives* of the model for a very broad class of effort technologies, and a relatively simple differential equation for the optimal trading policy of the large shareholder.
- d) *It generates important testable predictions.* The most important testable results of our paper include: a) Under rather general circumstances, stake of the large shareholder is likely to change over time towards a long-term equilibrium (competitive solution); b) Speed of adjustment is positively correlated with the stock price volatility; c) Speed of adjustment is negatively correlated with the degree of moral hazard and private benefits of control; d) Speed of adjustment is positively correlated with the presence of lockup restrictions, and negatively correlated with the amount of IPO underpricing. In addition, under conditions when the unique sub-game perfect equilibrium exists, company value is (weakly) increasing in shareholder's stake. This is consistent with hypotheses I and V about the relation between the corporate ownership and performance stated in the Section

1.1. We believe that our model should provide ample opportunity for much more probing time-series tests than previously performed.⁵⁹

e) *It connects traditionally separated branches of literature.* In this paper we attempted to bridge the gap between some traditionally separated branches of literature in asset pricing, corporate finance and corporate governance: asset pricing in the presence of a “large shareholder”; asset pricing in the presence of a moral hazard problem; optimal monitoring (corporate governance); and the principal-agent literature. Our model can describe the risk sharing/incentive alignment trade-off faced by a venture capitalist, or a large shareholder whether she is an insider who must expend effort to manage the firm or an outsider who expends effort to monitor the firm.

How robust are our conclusions to changes in model specification? Here are some issues that warrant further attention:

- Following the signaling argument of Leland and Pyle (1977), we expect that the information asymmetry between the agent and the insiders would decrease the speed of adjustment, *ceteris paribus*.⁶⁰ (Other forms of illiquidity, or the presence of transactions costs, would have a similar effect.) Incorporating dynamic effects of asymmetric information, and distinguishing them from the moral hazard effects considered here, would be an important next step in fully understanding the dynamics of large shareholder behavior.
- The possibility of a takeover could have a large impact on the adjustment policy. Indeed, if the large shareholder is a manager or manager-affiliated, selling too large a stake may jeopardize her position in the firm. Also, a large shareholder interested in changing the management would acquire, not sell, the firm’s stock. The threat of takeover would likely slow the speed of adjustment, at least for sufficiently low ownership stakes. In the future, it would be important to develop a model that would incorporate the possibility of a takeover into our dynamic framework.
- In addition to owning shares outright, large shareholder/managers often receive new shares through stock option grants. It seems intuitive that the presence of these stock option grants would offset insider selling and decrease the speed of adjustment. It would be an interesting question to explore how the introduction of non-linear contracts would change our results.
- Our model, for simplicity, assumes that there are no wealth effects and that future cash flows are independent of the past. This allows us to work in terms of certainty equivalents, and leads to a deterministic trading strategy for the large

⁵⁹ This could have important policy repercussions. Namely, imagine that a company is privatized using a “strategic partnership” strategy in which majority stake is given to a large shareholder in hopes that he would have the incentive to monitor, and, thus, improve the performance of the company. In a very volatile environment, according to our model, the agent would have the incentive to sell his stake or, in the absence of efficient public markets, strip the assets. Anecdotal evidence abounds on just such behavior in Russia, for example.

⁶⁰ Generally, the “lemons” problem leads holders of assets to retain a larger stake to signal high quality. See also Akerlof (1970) and Spence (1973) for the foundation of the asymmetric information and signaling models.

shareholder. Relaxing these assumptions would lead to a trading strategy that is contingent on stock price behavior. For example, with decreasing absolute risk aversion, the rate of sale would likely be negatively correlated with the stock's performance. These effects may be important empirically.

- In real life, companies may have more than one large shareholder and each of them may hold several risky assets in addition to their own company's stock. If they are precluded from colluding in their trading decisions, agents' trading policies will be interdependent and entail an important strategic component. In addition, including many risky assets into the analysis allows us to explore how the results in Admati et al (1994) generalize in the dynamic model. In a companion paper, Urošević (2001) explores these issues both theoretically and empirically.
- Regulations may prevent an agent from acquiring or disposing of the stake in a company. In any empirical study, therefore, effects of regulation must be considered.

Our model has left out several important issues. For example, there are likely to be multiple agency relationships associated with the firm. Huddart (1993) develops a model in which there are many shareholders monitoring a single manager, and both shareholders' monitoring and the manager's effort are costly. In this setting, there are multiple agents who influence (directly or indirectly) the cash flows of the firm. Urošević (2001) extends the model of this paper to allow for multiple "influential" shareholders, and explores the strategic interaction that results in a dynamic framework.

Implicit in the derivation of our results is the assumption that the identity of the large shareholder in the economy is constant. In other words, investors who purchase the shares of the company do not become large shareholders themselves. It would be interesting to relax this assumption in the future. Towards that end, it would be interesting to explore, building on Zwiebel (1995), strategic games between the large shareholders and the small block holders in an explicitly dynamic setting.

Finally, our model has assumed that all trades by the large shareholder are observable by investors. In reality, large shareholders are only required to report their trades ex-post. Thus, there is likely to be asymmetric information about the size of the trade. Vayanos (2000) considers the dynamic trading policy of a large shareholder in a Kyle (1985)-type model with noise traders. His analysis focuses on market microstructure issues, so he does not consider moral hazard – the large shareholder has no effect on the cash flows of the firm. It would be interesting to combine the framework of his paper with ours. Presumably, the ability to hide trades would increase the speed of adjustment of the large shareholder's portfolio, and might also allow for hidden accumulation of shares. Another interesting extension would be to allow negotiated "block trades" in which the large shareholder can acquire a significant stake at a premium that does not fully reflect the value of the increased monitoring services.

These and many other interesting issues await future research.

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